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A palaeomagnetic study of the Basal Gneiss Region and Jotun Nappe Complex, central–southern Scandinavian Caledonides

J. D. A. Piper and T. J. Poppleton
Geomagnetism Laboratory, Department of Earth Sciences, University of Liverpool, PO Box 147, Liverpool L69, 3BX, UK

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SUMMARY

In the southern sector of the Scandinavian Caledonides a metamorphic gneiss complex (the Basal Gneiss Region or BGR) underthrusts a nappe complex of Precambrian granulites (the Jotun Nappe) in an assemblage collectively recording orogenic events culminating in the Scandian (Upper Silurian and Lower Devonian) episode. Palaeomagnetic study of a large regional sample spanning the combined outcrop identifies properties strongly controlled by metamorphic retrogression to amphibolite and greenschist facies. Low blocking temperature (lbt) components in the BGR have E+ directions with a mean of $D = 69°$, $I = 60°$ ($N = 19$ sites, $\alpha_{95} = 7.0°$); sporadic higher blocking temperature components define a dual polarity northeast negative/southwest positive (NE−/SW+) axis with a mean of $D = 42°$, $I = -24°$ (20 sites, $\alpha_{95} = 8.1°$). Granulite facies rocks in the Jotun Nappe yield a high-quality palaeomagnetic record reflected in high Q_n values and dominated by medium and high blocking temperature (hbt) components. A dual polarity ('A') axis defines a palaeofield migration from NE+ to E+ during uplift and cooling of this terrane with a predominance of normal polarities. A smaller population of lbt ('B') components show E+ directions distributed along the Mesozoic palaeofield migration path for Eurasia, and probably acquired during brittle tectonic events of this age. 'A' magnetizations from 86 sites define mean south pole positions between 301°E, 2°N and 325°E, 10°S linked to uplift-related cooling following the climactic Scandian orogenic episode and dated at 420–400 Ma from the collective geological and radiometric evidence. Both the outward and return segments of an APW loop may be represented. The polar swathe correlates with (i) a higher blocking temperature component from the BGR, (ii) the uplift magnetization record in the western orthotectonic Caledonides of Scotland, and (iii) primary magnetizations from the Siluro–Devonian molasse facies in Britain. This study shows that the westerly extension of the APW path identified by British Siluro–Devonian data is not peculiar to that crustal segment but applies to continental Europe, and presumably to Laurentia, as well. The pattern of remanence acquisition within the Caledonides suggests contrasting zones of thermal and thermochemical remanence, with the latter probably linked to fluid migration consequent upon orogenic loading.

Key words: Basal Gneiss Region, Jotun Nappe, metamorphic rocks, palaeomagnetism, Phanerozoic, Scandinavian Caledonides.

1 INTRODUCTION

The magnetic record recovered from metamorphic terranes comprises an important part of the Precambrian palaeomagnetic database (e.g. McWilliams & Dunlop 1978; Morgan 1981; Piper 1981, 1985, 1987). Sedimentary and volcanic successions are a much more important component of the geological column in the Phanerozoic eon (<590 Ma), and there has not hitherto been a pressing need to study Phanerozoic metamorphic rocks. However, it is becoming increasingly evident that metamorphic rocks will be able to fill important gaps in the areal and temporal coverage of the palaeomagnetic record. Metamorphic terranes may record the geomagnetic field over long periods of time, during

uplift and cooling. Unfortunately they do not always possess a suitable magnetic mineralogy and grain size to preserve a fossil magnetization and the effect of metamorphism (especially at lower grades) is often to transfer the iron into the paramagnetic silicate minerals (Rumble 1978). Even if a stable magnetization has been imparted to the rock, it is generally by a chemical and/or mineralogical alteration rather than by a purely thermal effect: whilst theories linking the blocking temperature and intensity of magnetization to cooling rate may allow the rate and time of cooling through this temperature range to be estimated in the latter case (e.g. Dodson & McClelland-Brown 1980), this theory can probably only be realistically applied to very dry situations (to certain granulites for example) in the metamorphic environment. In most other situations chemical changes will take place during cooling; a positive link between magnetic components and the metamorphic history, as defined by metamorphic grade and isotopic closure in the metamorphic minerals, may then be difficult to establish. In favourable situations magnetic components can be linked to the deformational history by the application of standard palaeomagnetic field tests, although the overall orientation of the metamorphic complex at the time of magnetization cannot be directly determined.

In spite of these reservations, the palaeomagnetic record of metamorphic terranes may contribute towards unravelling aspects of their thermal history which would otherwise go undetected. They may, in addition, yield information on the scale of ancient apparent polar wander (APW), and thereby permit a comparison with short-term results from contemporaneous igneous and sedimentary rocks. This paper reports a large regional palaeomagnetic investigation of a Caledonian metamorphic terrane, illustrating the problems and potential of palaeomagnetic investigations of metamorphic rocks of Phanerozoic age.

The Caledonian orogenic belt in Britain has yielded much palaeomagnetic data relevant to the late Ordovician to Devonian interval. The magnetic imprint here has largely escaped the influence of comprehensive overprinting in Carboniferous–Permian times (Briden, Morris & Piper 1973; Briden, Turnell & Watts 1984; Trench, Bluck & Watts 1988) which has plagued many studies in rocks of this general age from elsewhere in Eurasia. Studies have been extended to the metamorphic sector of the orogen mainly by the work of Watts (1982, 1985). An investigation in the western sector of the Northern Highlands of Scotland (Watts 1982) illustrates a particular problem associated with metamorphic studies: a comprehensive magnetic imprint of Upper Silurian–Lower Devonian age appears to have gone entirely unrecorded in the K–Ar isotopic record; the metamorphic minerals record argon closure ages which are 10–60 Ma older (Dewey & Pankhurst 1970) than the age of magnetic blocking. The remanence also post-dates isothermal surfaces deformed by the local F3 episode of deformation. In contrast, the Barrovian zones of the Grampian Highlands (Watts 1985) define a remanence acquired between the regional F3 and F4 folding episodes.

2 OUTLINE GEOLOGY

The northeastward continuation of the Caledonian orogen into Scandinavia is composed predominantly of a thick nappe pile which was metamorphosed during and after emplacement following closure of the Iapetus suture. The igneous component in this sector of the orogen is almost entirely syn- and pre-tectonic in origin and unmetamorphosed sediments are confined to small basins linked to late Silurian and younger extensional and strike slip events (e.g. Torsvik *et al.* 1988). Caledonian palaeomagnetic investigations in this region must therefore concentrate largely on metamorphic rocks.

The metamorphic terranes comprise nappes thrust from west to east onto the Fennoscandian (Baltic) Precambrian platform in an orogenic event which culminated in Middle–Upper Silurian times (Roberts & Gee 1985), and followed closure of the Iapetus Ocean by late Ordovician times. The nappes are grouped into four major complexes comprising the outcrop from east to west of the Lower, Middle, Upper and Uppermost Allochthons (Fig. 1). The Lower Allochthon comprises a wedge of late Precambrian and early Palaeozoic sediments which has been detached and translated on a basal sole thrust traceable along the entire length of the orogen. The Middle Allochthon is comprised mostly of heavily mylonitized slices of Precambrian basement; it is metamorphosed to middle and upper greenschist facies and records much higher strains than those recorded in the Lower Allochthon. The Upper

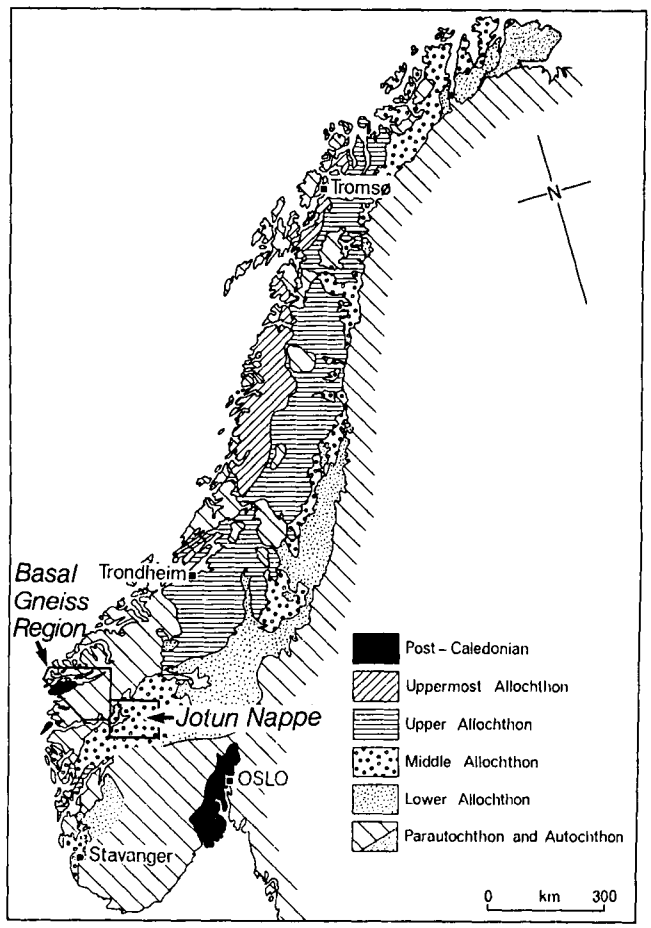


Figure 1. Regional tectonostratigraphic map of the Scandinavian Caledonides simplified after Roberts & Gee (1985) and showing the location of the Basal Gneiss Region and Jotun Nappe segments which form the subject of this paper.

Allochthon contains extensive evidence of igneous activity, with successions derived from a variety of destructive plate margin environments linked to the former site of the Iapetus Ocean. These rocks record deformations in early Ordovician (Finnmarkian) and Siluro–Devonian (Scandian) tectonic episodes and the metamorphism reached upper amphibolite grade or higher. The Uppermost Allochthon comprises nappes incorporating both metamorphic basement and a range of supracrustal rocks into which are emplaced a variety of synorogenic granites and subordinate gabbros. Nappe emplacement was succeeded by uplift, erosion and deposition in Lower and Middle Devonian times with molasse deposits accumulating in fault-controlled basins. These basins were, in turn, influenced by an episode of folding, thrusting and low-grade metamorphism in Upper Devonian and Lower Carboniferous times.

In central–southern Norway the overall tend of the orogen changes north to south from *circa* N30°E to N60°E and continues with this structural grain to the SW coast (Fig. 1). In the latter segment, nappes belonging to the Lower and Middle Allochthons (the latter including the Jotun Nappe of this study) are partially underthrust by an autochthonous or para-autochthonous metamorphic complex [the Basal (or western) Gneiss Region, subsequently referred to here as the BGR] belonging to the NW foreland of the orogen (Roberts & Gee 1985). Numerous age determinations demonstrate that rock units comprising the BGR were formed in mid-Precambrian times between *circa* 1750 and 1500 Ma (e.g. Brueckner 1979; Gebauer *et al.* 1985). The relative importance of Precambrian and Caledonian tectonism has long been disputed but it is now widely accepted following Holtendahl (1938), that the Precambrian rocks were strongly reworked by Caledonian orogenesis. The most important expression of this later deformation is the folding of Caledonian nappe sequences into the Proterozoic basement as large-scale recumbent folds. This deformation was responsible for resetting zircon U–Pb and some Rb–Sr whole rock systems (Lappin, Pidgeon & Van Breemen 1979). The polyphase ductilely deformed rocks have been metamorphosed to epidote–amphibolite grade with smaller zones of granulite grade. The contact with the overlying Jotun Nappe is a major tectonic discontinuity representing the site of substantial crustal contraction (Cuthbert, Harvey & Carswell 1983). The influence of Caledonian rejuvenation of the older Precambrian basement increases towards the west and northwest where the (high P–T) eclogite facies of metamorphism is widely represented in pods and lenses; these eclogites now show several stages of metamorphic retrogression and mineral assemblages constrain the

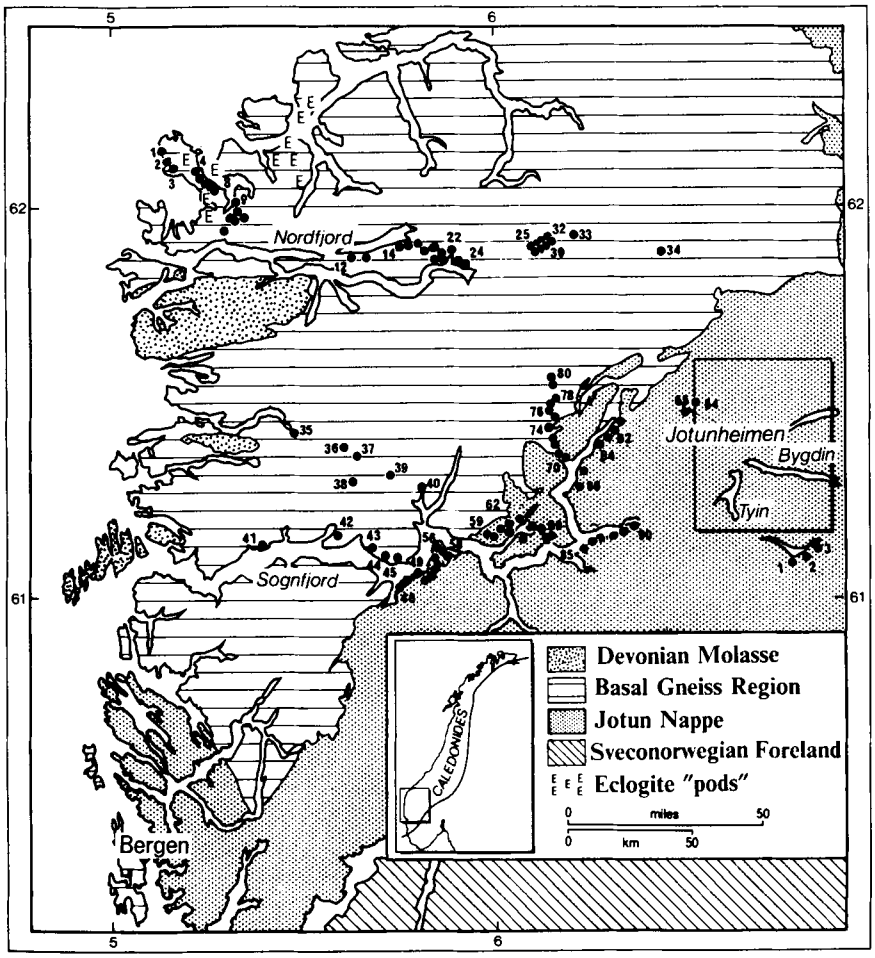


Figure 2. Outline geological map of the Basal Gneiss Region and Jotun Nappe showing the locations of palaeomagnetic sampling sites. The boxed area is the sampled section of the Jotunheimen Massif shown in expanded form in Fig. 3.

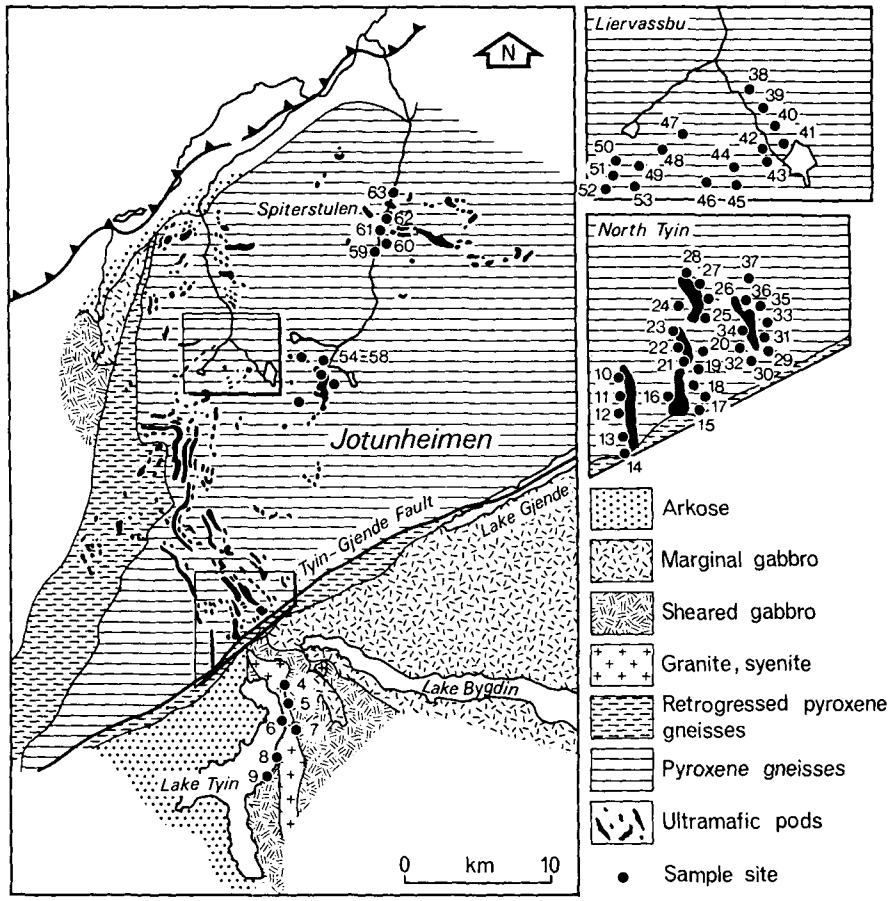


Figure 3. Geological map of the Jotunheimen Massif sector of the Jotun Nappe with geological outcrops simplified after Battey & McRitchie (1973); the palaeomagnetic sampling sites are also shown.

probable conditions of Caledonian regional metamorphism (Griffin & Brueckner 1985): in the vicinity of Sognfjord a maximum temperature of 500 °C was associated with a pressure of 10–12 kbar; metamorphic conditions increased (at presently exposed levels) to about 800 °C at a pressure of 18–20 kbar in the NW. This is interpreted as a metamorphic gradient imposed when the BGR was over-ridden from the NW by the Laurentian plate. The continent–continent collision culminated in the climactic events of the Caledonian orogeny in mid-Ordovician to early Silurian times (Griffin *et al.* 1985). The Jotun Nappe was emplaced during at least five phases of Caledonian deformation (Twist 1985); retrogressive metamorphism during subsequent uplift and cooling is recorded by local conversion of granulites to amphibolites.

The Jotun Nappe was emplaced onto the Fennoscandian foreland with a contact defined by a basal mylonite zone (Bryhni & Sturt 1985). It is composed predominantly of a *circa* 5 km thick layer of plutonic rocks of mid-Precambrian age (Scharer 1980). The study areas include the basal part of the nappe exposed in sections along Sognfjord also incorporating anorthositic rocks (Fig. 2), and the highest topographic sector of the nappe exposed between 1 and 2 km above sea level in the Jotunheimen Massif north of Lake Tyin (Fig. 3); this nappe therefore affords a useful opportunity to test the response to uplift-related cooling. Country rocks in the latter area are coarse granulites, locally retrogressed to amphibolites, into which have been emplaced a range of basic and ultrabasic intrusions (Battey & McRitchie 1973).

3 PALAEOMAGNETIC SAMPLING

Rock units belonging to the Basal Gneiss region were sampled at 34 sites (numbered 1–34 in Fig. 2) in the Nordfjord area near the central part of the BGR outcrop. A further 38 sites were sampled in the Sognfjord area *circa* 100 km further to the south (numbered 35–58 and 70–84 in Fig. 2). The total sample comprises 800 cores derived from a range of syn- and pre-Caledonian intrusions of basic to intermediate composition and a range of metasedimentary material; 58 samples come from eclogite pods (Fig. 2).

Sampling in the basal part of the Jotun Nappe was concentrated on the anorthositic facies exposed near sea level along Sognfjord (sites numbered 60–68 and 85–90 in Fig. 2) but included two amphibolitized basic intrusions (59, 69) which yielded no useful palaeomagnetic information. Sampling of the high levels of the Jotun Nappe exposed in the Jotunheimen Massif was concentrated on granulites and basic to ultrabasic intrusions emplaced into them (sites numbered 10–64 in Fig. 3). A major ENE–WSW tending fault [the Tyin (or Laerdal)–Gjende Fault] cuts the central part of the Jotun Nappe and down-throws a suite of lower grade rocks against the

granulite massif in the study areas; nine sampling sites (1 to 3 SE of Lake Tyin in Fig. 2 and 4–9 in Fig. 3) are located in rocks in this downthrown block.

All samples were collected in the form of cores drilled with a portable motorized drill from sample sites typically distributed over several metres of outcrop. Orientations were made by sights on the Sun and/or magnetic bearings, in conjunction with topographic sighting. Cores were sliced in the laboratory into 2.4 cm length cylinders. They were then subjected to progressive thermal demagnetization in steps of 50 or 100 °C, reduced to 20 °C steps at higher stages of treatment approaching the Curie points of the magnetic carriers, or to progressive alternating field (AF) cleaning in incremented steps of 5 or 10 mT to peak fields of up to 140 mT. Selected core ends were used for thermomagnetic, isothermal remanent magnetization (IRM) and low-temperature susceptibility (k) investigations.

4 PALAEOMAGNETIC RESULTS

4.1 Magnetic properties

Amphibolite or retrogressed greenschist facies material predominates in the collection of metasediments and intrusions from the BGR and shows a preponderance of low Q_n (the ratio of remanent to induced magnetization) values (Fig. 4).

Intensities of remanent magnetization before demagnetization (J_0) range over four orders of magnitude ($J_0 = circa\ 10^4$–10^7 A m^{2}kg^{-1}) and susceptibilities (k) over more than three orders of magnitude ($k = 2 \times 10^3$ – 10^6 SI units). There is no clear distinction between the J_0–J_I fields defined by the different rock types sampled (amphibolites, gabbros, gneisses, dunites and eclogites). In the event, and in general conformity with theoretical prediction (Stacey 1963), only the samples with higher Q_n values (all >0.1) yielded significant vectors (defined here by the mean direction for three or more successive demagnetization steps yielding a cone of 95 per cent confidence of less than 20°, with selected components checked independently by application of LINEFIND). An alternative technique used in this study for recognizing material able to yield useful directional information is low-temperature susceptibility analysis. Magnetically unstable samples show either a progressive increase in susceptibility between liquid nitrogen (−197 °C) and room temperatures, of little or no change (Fig. 5a). The patterns show more variation in shape than the Type 1 curves reported by Senenayake & McElhinny (1981) from young basalts, but the presence of multidomain magnetite carriers or a preponderance of paramagnetic phases would accord with the magnetic characteristics of this group. The bulk of the investigated samples yielding useful palaeomagnetic information define a peak in the low-temperature susceptibility curve between −180° and −140 °C (Fig. 5), correlating with the change in sign of the K_1 anisotropy constant in magnetite and compatible with the Type 2 behaviour of Senenayake & McElhinny (1981); the stability of remanence is attributed to grain sub-division by exsolution of a Ti-rich mineral. Thermomagnetic charac-

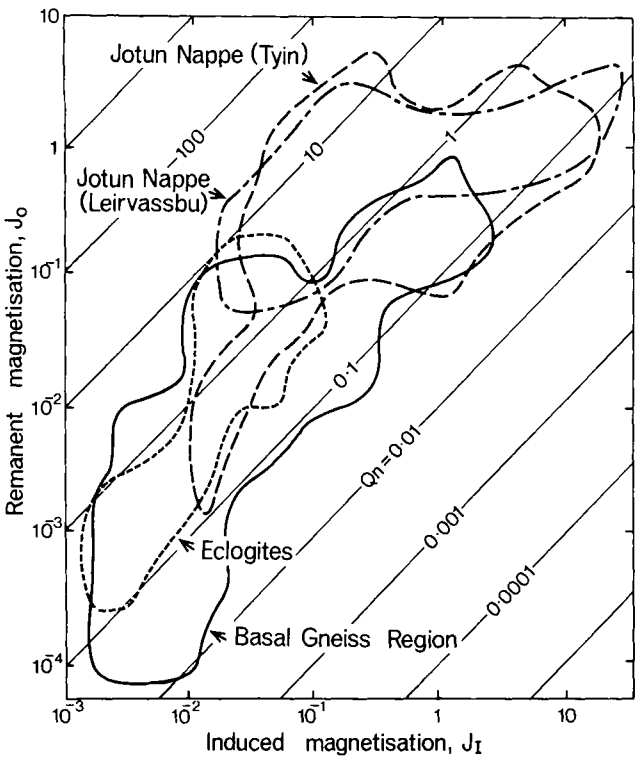


Figure 4. Induced magnetization, J_I, versus remanent magnetization, J_0, fields defined by 497 samples from the Basal Gneiss Region (BRG), 57 samples from eclogite pods in the western part of this division, 186 samples from the southern sector of the Jotun Nappe in the Tyin area and 34 samples from the central–northern sector of the Jotunheimen Massif (Leirvassbu area). ($J_I = kF$, where F is the present-day geomagnetic field intensity at the sampling site.)

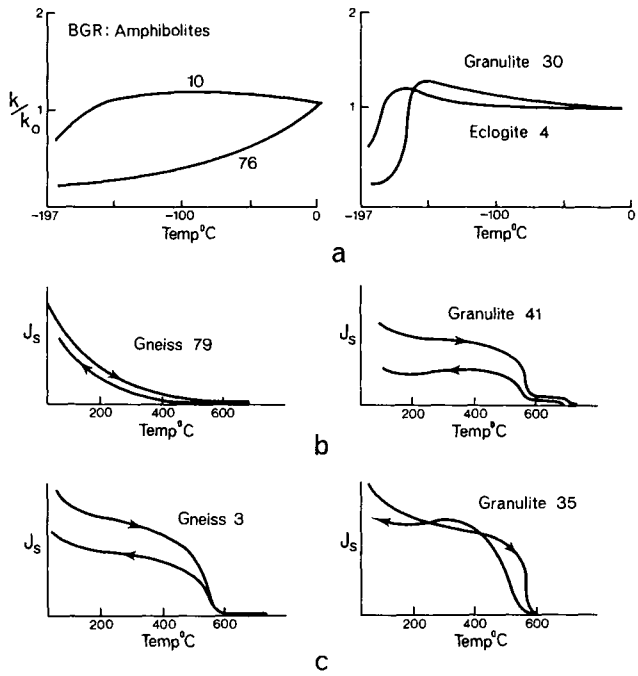


Figure 5. Low-temperature susceptibility characteristics (a) and typical thermomagnetic determinations (b) and (c) from palaeomagnetic samples in the Basal Gneiss Region (left) and Jotun Nappe (right). The susceptibility values are normalized to the value (k_0) at room temperature and the thermomagnetic curves are shown as saturation remanence (J_S, arbitrary value) versus temperature.

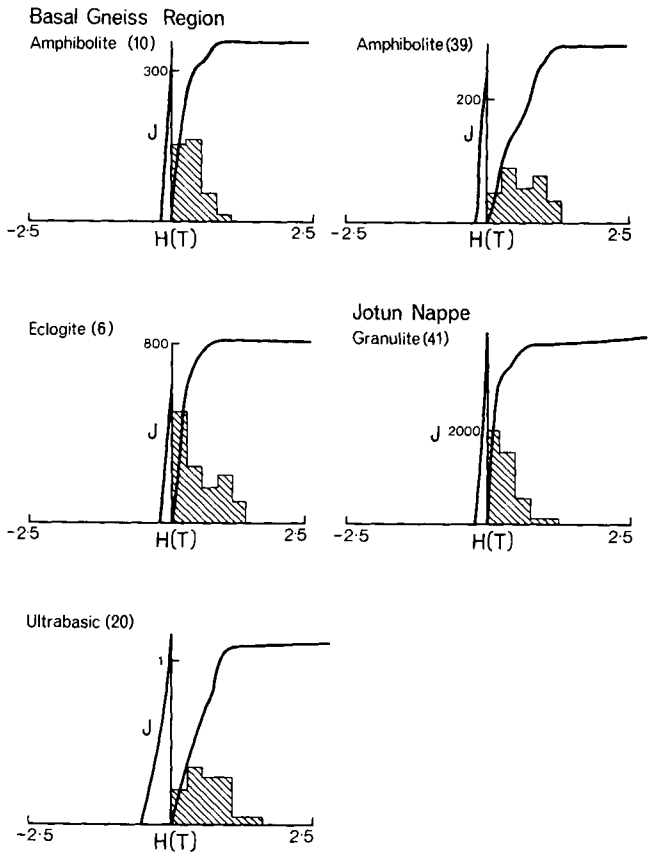


Figure 6. Some IRM acquisition characteristics of metamorphic samples from the Basal Gneiss Region and Jotun Nappe.

teristics are of little general use for identifying suitable palaeomagnetic material but of some value for identifying magnetic carriers. Examples from metamorphic material in the lower grades of metamorphism show asymptotic decrease in saturation remanence (J_S) with increasing temperature [sample 79 in Fig. 5(b)] linked to a predominance of paramagnetic carriers obeying the Curie Law. Other (usually stable) samples show a discrete magnetite Curie point [samples 3, 35 and 41 in Fig. 5(b) and (c)]. Separation of heating and cooling curves is indicative of magnetic carriers in a metastable condition and alteration during heating in air. Such samples cannot generally be effectively treated by thermal demagnetization. Discrete haematite Curie points [sample 41, Fig. 5(b)] are identified at only a few sites in the BGR but are prominent within the higher grade rocks of the Jotun Nappe, notably in the northern sector of the Jotunheimen granulite massif. Isothermal Remanent Magnetization (IRM) acquisition studies also identify a preponderance of magnetite and/or sulphide carriers in the bulk of this collection, which mostly saturate in fields below 0.5 T (Fig. 6, samples 6, 10, 20 and 39); a smaller fraction of (haematite-bearing) samples continue to show a small progressive increase in IRM above this (Fig. 6, sample 41).

The contrasting J_0/J_1 and corresponding Q_n, fields (Fig. 4) defined by rock suites comprising this collection are reflected in the responses to demagnetization. Fifty four per cent of the material from the BGR failed to yield significant vectors with progressive cleaning treatment, compared with only 6 per cent in the Jotun Nappe. In the stable fraction a much larger proportion have low blocking temperatures (56 per cent) in the BGR than in the Jotun Nappe (18 per cent). The high percentage of magnetically stable samples in the Jotun Nappe is reflected in the large proportion of samples (82 per cent) with medium to high blocking temperatures.

4.2 The Basal Gneiss Region

Since the magnetic record in metamorphic terranes is typically complex, it is seldom advisable, or even possible, to attempt to characterize individual remanence directions; more often remanences with common directional properties have varied magnetic properties indicative of contrasting carriers. Figs. 7(a) and (b) illustrates the predominance of components with intermediate to steep positive inclinations defined from thermal demagnetization. Some [e.g. samples 16-6(a), 49-6 and 75-4(b) in Fig. 7] have a steep inclination indistinguishable from the present geomagnetic field direction although the declinations are variable [4-4(a), 40-9 and 48-2(b) in Fig. 7]. The majority of these samples shows only a single significant component of magnetization, which survives the Curie point of magnetite (550 °C) or haematite (675 °C) (samples from sites 75 and 16 respectively) with distributed blocking temperature spectra. In other samples the inclination is shallower than that of the present field (e.g. samples 9-6, 11-1, 14-2, 18-7, 25-6 and 26-4) but the declination is consistently to the NE. Remanence components fall into at least two groups with a distinction recognized between vectors with a NE declination and positive inclination surviving to the Curie point of magnetite, and components of similar direction but with lower and more distributed blocking temperatures; the latter are sometimes removed at temperatures below the Curie point to isolate more easterly components also with positive inclinations, but with narrower and higher blocking temperature spectra [e.g. sample 26-4 in Fig. 7(b)]. Two samples from site 18 illustrate this dichotomy of behaviour with only the E+ component present in sample 5 but a NE component first removed below 550 °C in sample 7. In a minority of samples the NE+ or E+ components are further removed to recover a higher blocking temperature NE negative inclination component [sample 14-2 in Fig. 7(a)]. This direction is present as a discrete component in some samples (e.g. 8-13) and an antiparallel equivalent is also found (e.g. in 83-12). In a few samples the E+ or NE+ components are removed to recover a southerly shallow remanence with higher blocking temperatures [e.g. 10-2 in Fig. 7(a)]. Eclogite pods (samples included in sites 1–9) at the western extremity of the study area at Stadlandet (Lappin 1966, 1974) were sampled in anticipation that high metamorphic grades would favour retention of a stable remanence. In the event, as results shown in Fig. 8 and Table 1 indicate, the record is only patchily stable and complex in detail. Sometimes for sets of adjacent or closely spaced cores, some may be stable (notably at the centre of the pods) and others unstable. Thin sections show partial alteration producing ramifying veins of amphibole and chlorite, and we infer that retrogression has been responsible for transferring much of the iron into hydrous silicate phases. Only 34 per cent of this collection yielded magnetic components passing our acceptability criteria.

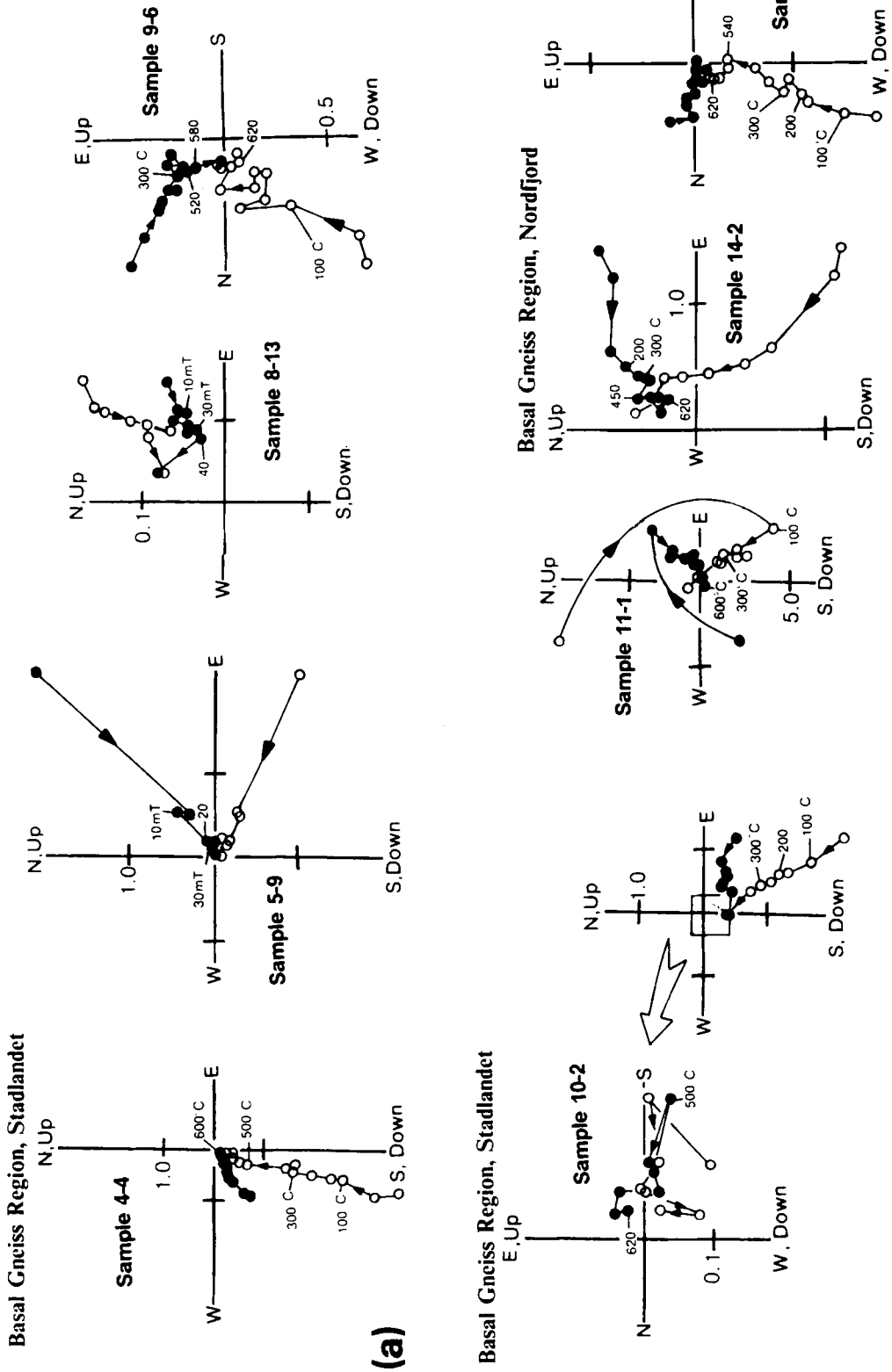


Figure 7. Examples of alternating field and thermal demagnetization of metamorphic rocks from the Basal Gneiss Region in its northwestern (a) and southeastern (b) sectors represented as orthogonal projections. Projections onto the vertical plane as open circles and projections onto the horizontal plane as closed circles; intensity scales are graduated in 10^{-5} A m^{2}kg^{-1} units. The site numbers refer to the sites shown in Fig. 2 and listed in Table 1.

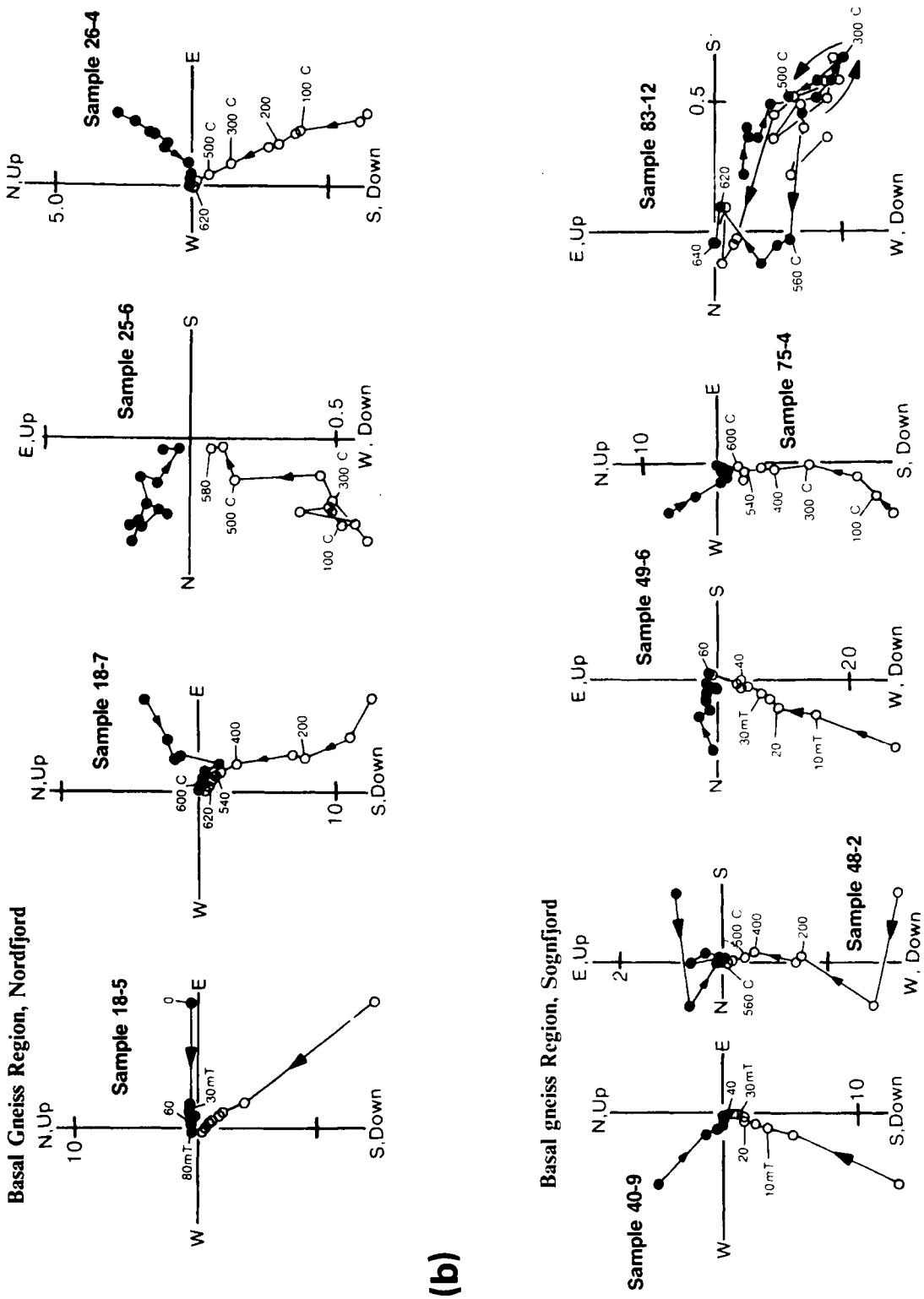


Figure 7. (*continued*)

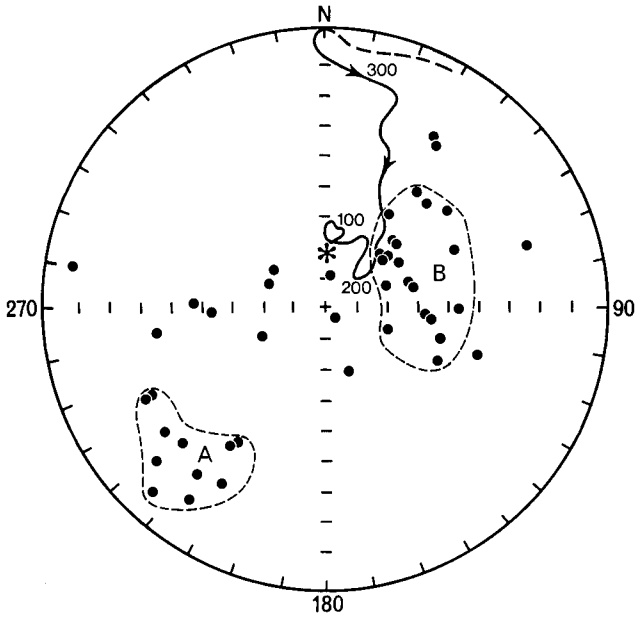


Figure 8. Site mean directions of magnetization from the BGR together with the palaeofield migration path for this region between 310 Ma and the present (asterisk) predicted from the APW path of the Eurasian plate (Piper 1987). Solid symbols are lower hemisphere projections and open symbols are upper hemisphere projections. The groupings shown as 'A' and 'B' in this figure define the directions employed in the mean calculations of the NE−/SW+ and N+ directions respectively listed in Table 3; the components defining the latter direction are shown with a common reversed (SW+) polarity. Equal area projection.

Table 1. Site mean palaeomagnetic results, Basal Gneiss Region, south-central Scandinavian Caledonides.

Site No.	Rock type	Component Character-istic*	N	D	I	k	α_{95}
(i) Nordfjordur							
1	G	TBm	2	247	69	38	-
3	G	TB	4	185	47	14	26
5	E	TB	3	30	-24+	19	29
6	E	TBl-m	5	184	46	18	18
7 (i)	E	TB	4	34	26+	8	35
(ii)		TBl-m	2	35	-14+	20	-
8	GD	TB	2	61	69+	402	-
9	E	TBm	3	52	-26+	18	30
10	A	TB	3	114	50+	28	24
11	A	TBm	3	65	44+	62	16
13	G	TB	3	244	26+	13	36
14 (i)	A	TBm-h	2	46	-27+	356	-
(ii)		-	3	132	86	21	28
18	A	TB	6	49	65+	20	15
19	A	TB	3	4	79	162	10
20	A	TB	2	48	59+	59	-
26	G	TBl-m	8	58	61+	103	5
28	G	TB	3	48	64+	31	23
29	G	-	4	92	57+	7	39
30	G	TB	3	74	61+	58	16
32	GR	TBm	4	46	66+	68	11
33	A	TB	5	109	69+	27	15
(ii) Sognfjordur							
35	A	TB	4	94	56+	39	15
37	A	TB	4	44	42+	20	21
38	A	TB	2	34	53+	853	-
39	A	TBl-m	3	213	39+	13	36
40	A	TBm	3	268	53	11	40
41	G	TBm	2	46	68+	28	-
42 (i)	A	TB	3	51	39+	19	29
(ii)		TBm	2	43	-9+	8	-
43	A	TBh	3	72	22	73	15
48	A	TBm	4	157	69	19	22
52	GD	TBm	3	313	57	19	29
54	A	-	3	287	-39	7	51
55	GD	TBm	4	262	35	5	45
56 (i)	AN	TBm-h	3	34	23	20	28
(ii)		-	3	34	-37+	14	34
58	A	TBm	4	113	52+	12	27
71	A	TB	2	228	17+	51	-
73	A	TB	2	272	48	73	-
74	A	TB	2	307	69	50	-
75 (i)	A	TBm	3	294	70	34	21
(ii)		-	3	326	-7	31	22
76	A		2	84	-10	28	-
77	G+A	TB	2	71	62+	24	-
78 (i)	U	-	2	280	7	28	-
(ii)		TB	3	90	46+	11	40
83	A	TBl	4	217	22+	49	13

Footnote: D and I are the mean declination and inclination derived from N samples with Fisher precision estimate k(=(N-1)/(N-R) and semi angle of cone of confidence of α_{95} about the mean direction.
*Blocking temperature characteristics as defined from thermally-demagnetised samples : low blocking temperature (TBl) = unblocked between 0 and 500°C, and medium blocking temperature (TBm) = unblocked between 500 and 580°C, and high blocking temperature (TBh) = unblocked between 580 and 680°C. Rock types are abreviated in this Table and Table 2 as G, Gneiss; E, Eclogite; GR, Granite, granodiorite etc; A, Amphibolite; G, D, Gabbro or diorite intrusion; AN, Anorthosite; U, ultrabasic intrusion, and GA, Granulite + components employed in group mean calculations.

On the basis of the thermally demagnetized data set, components are classified as being low blocking temperature if removed by 500 °C, intermediate blocking temperature if removed by the Curie point of magnetite (580 °C), and high blocking temperature if they survive into the blocking temperature range exclusive to haematite (580–680 °C). Employing this scheme the majority of the remanence components (56 per cent) identified in the BGR has low blocking temperature and a further 36 per cent has medium blocking temperature; they include all but one of the E+ site mean components (Fig. 8). Little evidence for a record resident in haemo-ilmenite phases has been recognized in this terrane, although some members of the subsidiary group of NE−/SW+ vectors have properties qualifying for this group and might therefore, have recorded an overprint resulting from oxidation of primary phases.

In principle, single palaeomagnetic samples from metamorphic terranes may be expected to yield time-averages of the ancient geomagnetic field and therefore palaeomagnetic poles rather than virtual geomagnetic poles. No reversals of magnetization have been identified within individual cores however; hence time intervals represented by the critical blocking temperature ranges are not clear. Accordingly site mean directions in Table 1 are calculated from groupings recognized between two or more cores at single sites; these means are used to calculate the group means for the N–NE+ components and the minority NE−/SW+ axis in Table 3.

Collectively the BGR illustrates a predominance of E–NE directed positive magnetizations defined by both thermal and AF demagnetization and, with a few exceptions, unblocked below the Curie point of magnetite. This mineral (together possibly with pyrrhotite in some cases) is therefore the likely carrier of the remanence as indicated by the ancillary investigations noted in Section 4.1. The direction of this predominant remanence, together with its distributed range of unblocking temperatures, suggests that it may record a Mesozoic overprinting event. Magnetic overprinting linked to development of the North Sea graben has been identified at several places around this basin (e.g. Turner, Vaughan & Tarling 1978; Storetvedt *et al.* 1978; Løvlie & Mitchell 1982; Storetvedt & Torsvik 1983). Whilst Mesozoic

and Cainozoic overprinting is the probable explanation for at least some of the components included in the 'B' group, it is not an entirely satisfactory explanation for two reasons. First, these vectors plot systematically to the east (in some cases albeit only marginally so) of the palaeofield migration path predicted from the APW path for the Eurasian Plate (Fig. 8). Second, some have a complex origin and the easterly positive component in the vicinity of the Mesozoic field vector is resident in higher coercivity/blocking temperature fractions. Since the sampling sites are mostly remote from the coastal margin, a simple tectonic flexure cannot explain this discrepancy. Furthermore the sites classified as Group 'B' in Fig. 8 have a range of declinations (85°) appreciably greater than the range of inclination (35°). This effect is a possible signature of local tectonic rotations and suggests that these components may, in part, pre-date some of the tectonism within the BGR. These observations, together with the similarity of this direction to components observed elsewhere in the Scandinavia Caledonides (Poppleton & Piper 1989; Piper *et al.* 1990) and in the Unst ophiolite of Shetland (Taylor 1988) suggest that some 'B' components might also be interpreted in terms of the Caledonian history of this region. In the absence of diagnostic field tests, further interpretation of these components is not yet warranted.

The NE−/SW+ components define a subordinate field axis remote from the post-310 Ma palaeofield migration path for the area (Fig. 8) and more readily interpreted in the context of Caledonian history of this region and the overlying Jotun Nappe (Section 5).

4.3 The Jotun Nappe

In general, rock suites in the Jotun Nappe prove to be much more suitable for palaeomagnetic investigation than those within the adjoining BGR. This is reflected in a higher range of NRM intensities ($J_0 = 10^{-3}$–10^{7} A m^{2}kg^{-1}) and susceptibilities ($k = 4 \times 10^{-1}$–1×10^{-4} SI units), and in higher average Q_n ratios (Fig. 4). Peaked low-temperature susceptibility profiles are typical and all of 25 thermomagnetic determinations in granulite facies rocks defined discrete magnetite Curie points, with an additional 12 defining both magnetite and haematite Curie points (Section 3.1.)

There are numerous examples of probable Recent viscous components in the Jotun Nappe samples; they have variable declinations but uniformly steep positive inclinations [e.g. samples 10-3, 15-4, 34-7, 34-8 in Fig. 9(a) and (b) and samples 38-5, 89-3 and 90-1 in Fig. 10]. These components are always removed in the first few steps of treatment and, since these samples also carry more significant components with higher blocking temperatures and coercivities, they are not considered further here. The behaviour of these samples contrasts with that of a second group which also yield components with intermediate to steep positive inclinations but which survive to the Curie point of magnetite [e.g. samples 8-8, 20-4, 21-10, 23-2, 27-6 and 30-10 in Figs 9(a) and (b) and 88-8 in Fig. 10] or haematite [13-8 in Fig. 9(a) and 65-6 in Fig. 10]. In these cases a geological event rather than a thermal relaxation mechanism seems likely to be the cause. This inference is supported by the observations that the declinations are mostly northeasterly and inclinations are generally shallower than that of the present Earth's geomagnetic field. Only normally magnetized examples of this remanence have been found, and the correlation of these vectors with the Mesozoic palaeofield is noted below.

The most common characteristic remanence has E–NE declination and shallow negative inclination. It is sometimes discrete [e.g. 16-6 in Fig. 9(a)] but is mostly isolated following removal of the viscous component noted above. This component is efficiently subtracted by AF treatment [e.g. 9-9, 15-1 and 22-5 in Figs (a) and (b) and 61-1 in Fig. 10] and is evidently resident in the magnetite (Section 4.1). Nevertheless the blocking temperature spectrum is remarkably narrow [e.g. 20-4 in Fig. 9(a) and 60-10 in Fig. 10] and is mostly concentrated in the range 570–610 °C. For this reason the AF treatment is actually more effective at defining the vectors and in some cases it has succeeded in identifying a reversal within single samples (e.g. 40-3 and 42-9 in Fig. 10): a NE negative component is defined by the lower portion of the coercivity spectrum and a SW positive component is defined by the higher portion. In a few cores the NE negative component is only recovered from a residual haematite-resident component following removal of a NE positive component resident in the magnetite (e.g. 49-4 in Fig. 10).

This is the only characteristic remanence within the Jotun Nappe for which both polarities are recognized. A smaller fraction of the samples have SW positive components with comparable properties to the NE negative components. These SW components are progressively subtracted by AF cleaning and are therefore most probably magnetite-held [Section 4.1 and 10-3 in Fig. 9(a) and 89-3 in Fig. 10]. Blocking temperature ranges are also narrow, with the larger part subtracted between 570° and 590 °C [24-4 in Fig. 9(b) and 89-6 in Fig. 10]; in these cases AF treatment again defines the component structure rather better than thermal treatment (this point is illustrated by the two samples from site 89 in Fig. 10 as well as for the two samples from site 15 belonging to the NE negative group in Fig. 9). At six sites opposite polarities are identified at adjacent [e.g. 34-7 and 34-8 in Fig. 9(b)] or nearby cores. The SW positive components in the examples identified in this study appear to have somewhat higher coercivities [site 15 in Fig. 9(a) and sites 40 and 42 in Fig. 10)] than the more common NE negative components; if the coercivity spectrum is an approximate reflection of the blocking temperature spectrum in these rocks, this could record remanence acquisition in a reversely magnetized field and then in a normally magnetized field during decline of regional metamorphic temperatures through the blocking temperature spectrum. The NE−/SW+ components contrast with the NE+ components in that the latter always have distributed blocking temperature spectra [cf. 8-8 and 17-4 in Fig. 9(a)].

A total of 94 per cent of the samples treated by AF or thermal demagnetization yielded components passing our acceptability criteria. Site mean statistics for coherent direction(s) represented in two or more cores from the same site are listed in Table 2.

A small number of samples from the anorthosites exposed in the Sognfjord area have broadly overlapping blocking temperature/coercivity spectra and illustrate great circle

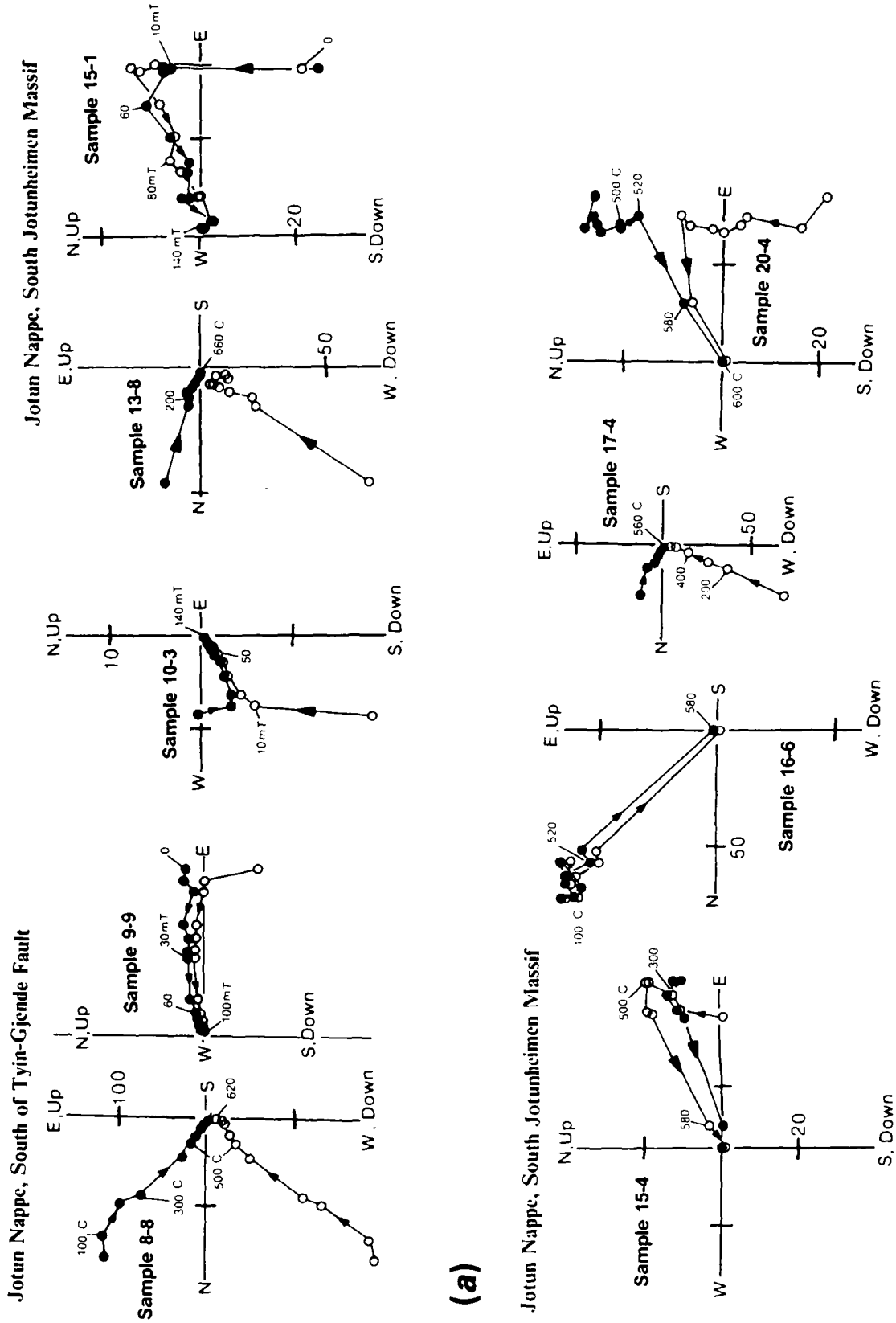


Figure 9. Examples of thermal and alternating field demagnetization of metamorphic rocks from the southern sector of the Jotun Nappe. Symbols etc. are as for Fig. 7 and the site numbers refer to those shown in Fig. 3 and Table 2.

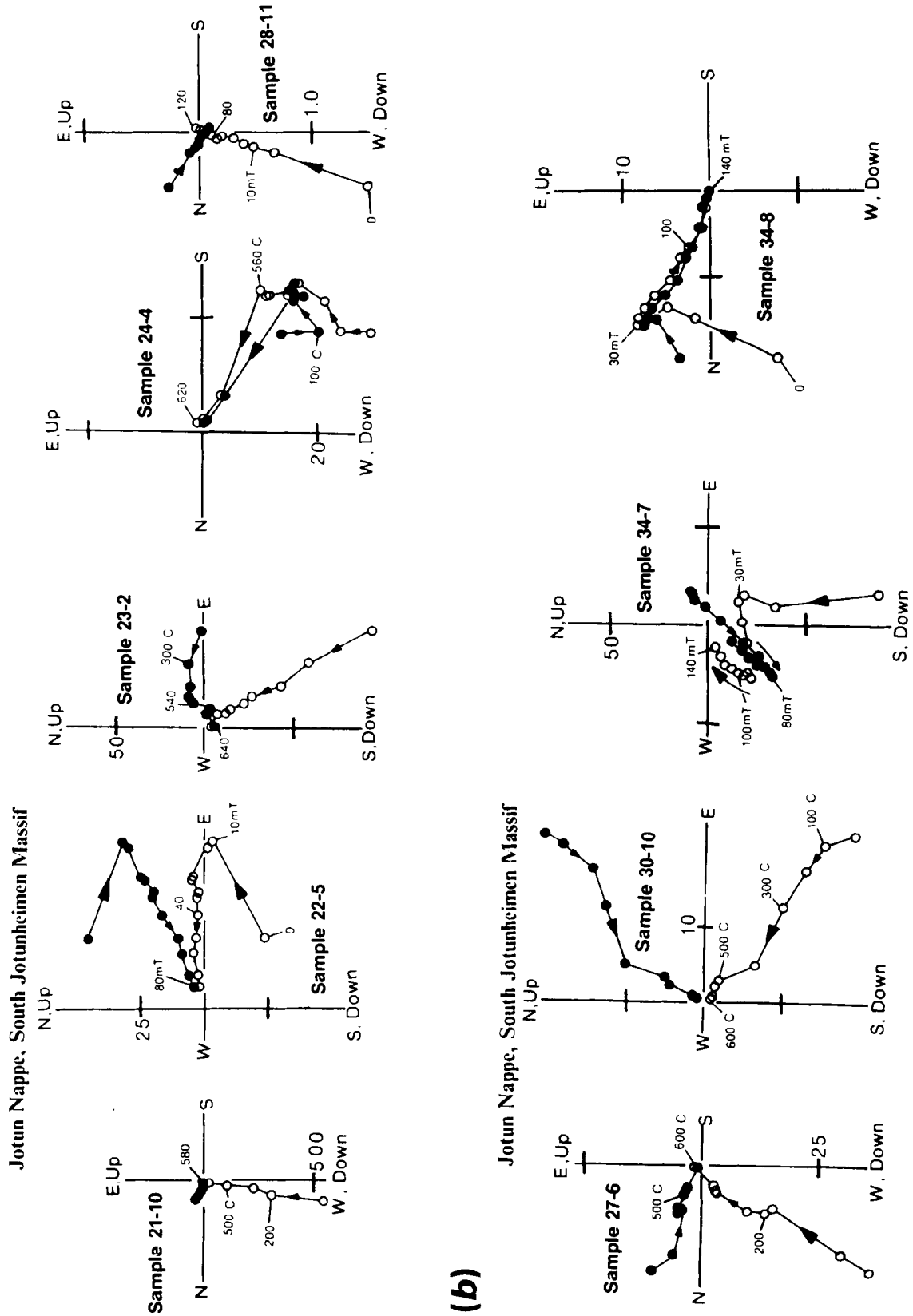


Figure 9. (*continued*)

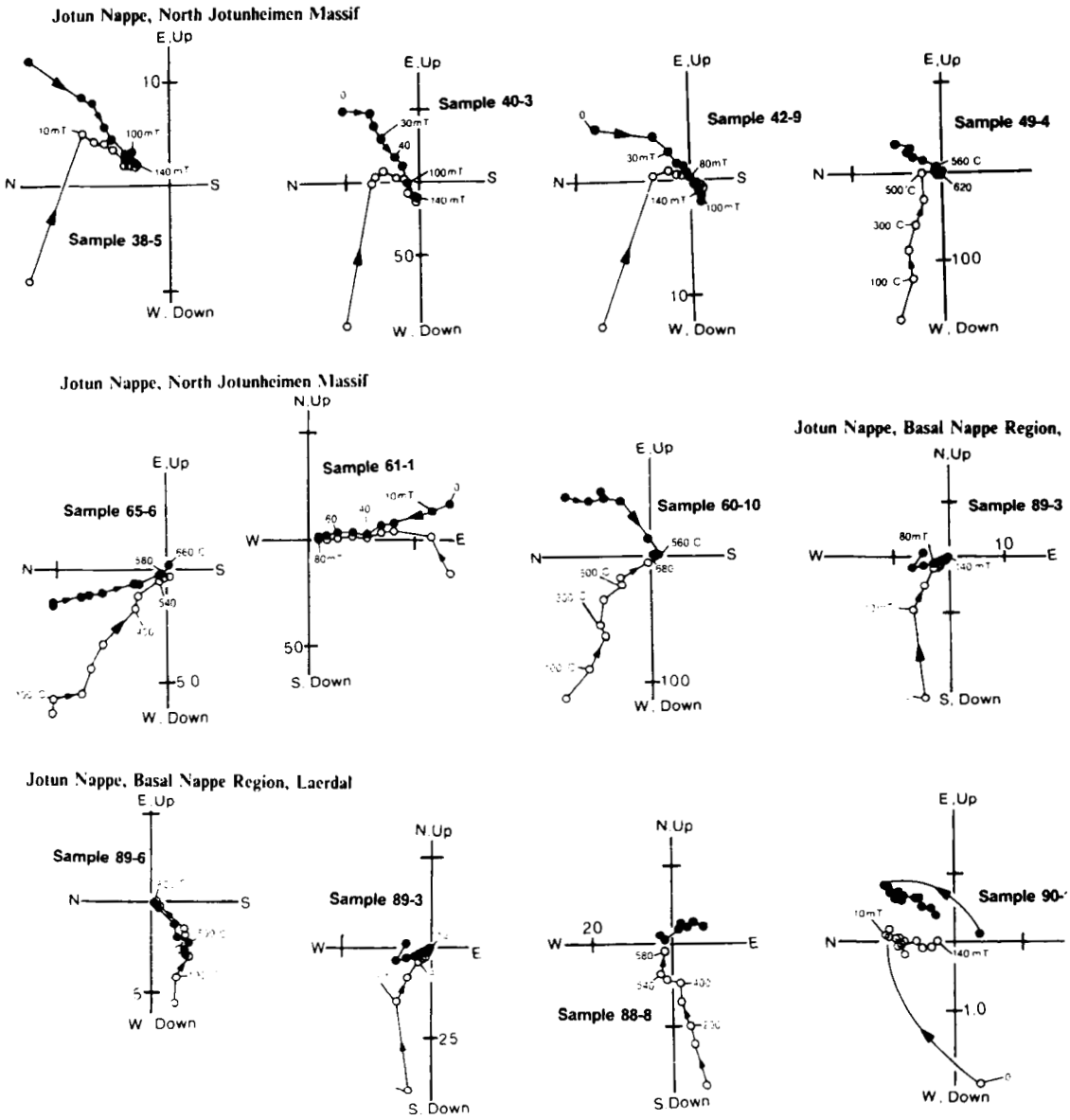


Figure 10. Examples of thermal and alternating field demagnetization of metamorphic rocks from the northern sector and basal regions of the Jotun Nappe. Symbols etc. are as for Figs 7 and 9.

migrations with progressive cleaning treatment (Fig. 11). The reason for dispersion of these trajectories is uncertain; it could result from the recording of several short term reversals of the magnetic field as these rocks cooled through the spectra recovered by demagnetization. The arcs converge to yield a graphical best fit direction of $D = 36°$, $I = -10°$ and an equivalent pole position at 328°E, 18°S (cf. Table 3).

The majority population of NE−/SW+ components is distinct from the predicted Carboniferous and younger palaeofield migration path for the area (Fig. 12b) and we therefore interpret this population as most probably representing the magnetic imprint acquired during uplift and cooling of the orogen. There are several characteristics which point towards this interpretation by implying a protracted acquisition during orogenic denudation as follows.

(i) Both host country rock granulites (or their retrogressed equivalents) and the large (anorthosite) and smaller (ultrabasic) intrusions possess essentially the same magnetic signature (Fig. 12c). Magnetic inversions are recorded within some individual sites and sometimes within single samples. We have identified no clear distinction between the imprint of the separate lithologies on a large scale other than between the base of the Jotun Nappe [where reversely magnetized rocks are more strongly represented, see Fig. 13(a)] and the higher levels. On a small scale, such as near the southern margin of the Jotunheimen Massif (Fig. 3 and Table 2), local differences are attributable to small differences in blocking temperature spectra and cannot be related to contact chilling. This is compatible with the plutonic history of these rocks: the metamorphic fabric, which is often present in the contact zones, records the common metamorphism of the granulites

Table 2. Site mean palaeomagnetic results, Jotun Nappe, south-central Scandinavian Caledonides.

Site	No*	Rock type	Component Character-istic **	N	D	I	k	α_{95}
(i) South of Tyin-Gjende Fault								
1		GD	TBh	4	73	-19+	7	37
2	(i)	GD	TBh	3	278	-41	9	-
	(ii)			2	183	-42	23	-
4		A	TBl	4	74	-28+	43	14
6		A	TBm	6	53	-24+	10	23
8		G	TBm	3	58	31	9	43
9		A	TBm	9	79	-3	42	8
(ii) Jotunheimen Massif, southern sector								
10		U	TBm	8	243	35+	18	13
11		GA	TBm	5	19	48	37	13
12	(i)	GA	TBm	9	38	-11+	8	20
	(ii)		TBl	4	28	42	136	8
13	*(i)*	*GA*	*TBm*	*6*	*40*	*-35*	*70*	*8*
	(ii)		TBl-m	2	29	38	94	26
14		A	TBl-m	8	214	-43	29	11
15		U	TBm	9	75	-20+	53	7
16	(i)	U	TBm	6	48	-19+	26	13
	(ii)		TBl-m	2	106	59	83	-
17	(i)	GA	TB	3	320	75	18	30
	(ii)		TBl-m	3	30	64	89	13
19	(i)	U	TBm-h	2	40	60	29	-
	(ii)		TBh	6	288	28	12	20
20	(i)	U	TBm	6	65	-18+	15	18
	(ii)		TBl	3	30	27	31	22
21	(i)	GA	TBl-m	4	43	72	82	10
	(ii)		TBm	4	58	-7	15	24
22		U	TBm	9	57	-1	77	6
23	(i)	U	-	3	63	-31+	68	15
	(ii)	U	TBm	4	69	49	36	15
24	(i)	U	TBh	9	216	7	17	13
	(ii)		TBm	5	238	35+	24	16
25	(i)	U	TBm	4	359	54	37	15
	(ii)		TBm	4	23	8	26	18
	(iii)		-	5	97	-52	26	15
26	(i)	U	TBl-m	6	35	26	16	17
	(ii)		TBm	3	75	-34+	12	37
27	(i)	GD	TBm	3	40	-16+	10	40
	(ii)		TBl-m	4	33	29	14	26
28		U	TBl	6	63	69	65	8
29		GA	TBh	9	3	-7	36	9
30		U	TBh	10	11	-28	28	9
31		U	-	6	53	-21+	18	16
32	(i)	U	TBm	7	235	29+	11	19
	(ii)		-	2	23	53	537	-
33	(i)	GA	TBh	5	15	23	16	20
	(ii)		-	8	51	-15+	42	9
34		U	TBh	12	30	-11+	62	5
35	(i)	GA	TBm	4	33	-32+	40	15
	(ii)		-	3	345	-43	105	12
36		GA	TBm	7	48	-11+	13	18
37	(i)	GA	TBh	6	45	-18+	22	15
	(ii)		TBm	3	38	36	20	28
(iii) Jotunheimen Massif, northern sector								
38		GA	TBm	7	46	-24+	120	5
40		GA	TBm	8	58	-15+	92	6
41	(i)	GA	TBl-m	4	31	48	29	17
	(ii)		TBm	3	42	-24+	8	46
42		GA	TBm	8	59	-22+	44	8
43		U	TBm-h	10	50	-23+	44	7
44		GA	TBm	7	50	20	8	23
45		U	TBm	7	70	-16+	30	11
46		U	TBm	8	48	-26+	47	8
47		GA	TBm	7	43	-30+	50	9
48	(i)	GA	-	4	275	-1	48	13
	(ii)		TBm-h	4	251	37+	74	11
49	(i)	U	TBl	4	31	60	136	8
	(ii)		TBm-h	7	56	-20+	22	13
50		GA+U	TBm	9	50	-25+	38	8
51		U	TBm	8	69	-26+	16	14
52		U	TBm	8	54	-13+	24	11
53		GA	TBm	11	50	-17+	75	5
54		GA	TBm	6	63	-36+	12	20
55		U	TBm	7	48	-17+	30	11
56		U	TBm	9	52	-17+	56	7
57	(i)	GA	TBm	4	46	-19+	48	13
	(ii)		TBl	2	49	71	64	-
58		U	TBm	8	59	-22+	47	8
59		U	TBm	3	45	13	22	27
60	(i)	GA	TBl-m	3	53	37	28	24
	(ii)		TBm-h	2	58	-19+	22	-
61		U	TBm	6	73	-5	56	9
62		U	TBm	7	64	-23+	52	8
63		GA	TBl-m	3	45	81	36	21
64		GD	TBl	3	30	17	9	45

Table 2. (*continued*)

Site	No*	Rock type	Component Character-istic **	N	D	I	k	α_{95}
(iv) Basal Jotun Nappe, Sognfjord								
59		AN	TBm	5	32	-26+	20	17
60		AN	TBm-h	7	241	24+	39	10
61		AN	TBm	5	242	3	19	18
62		AN	-	6	60	-21+	24	14
63		AN	TBm	3	47	-28+	7	52
64		AN	TBm	6	71	-30+	11	21
65		AN	TBm	3	65	-17+	53	17
66		AN	-	4	50	-27+	8	35
67		AN	-	7	237	11+	17	15
(v) Basal Jotun Nappe, Laerdalsfjorden								
87	(i)	AN	TBl	3	42	14	43	19
	(ii)		TBm-h	3	52	-20+	72	15
88		*AN*	*-*	*3*	*279*	*-17*	*16*	*32*
89	(i)	AN	TBm	6	243	43+	61	9
	(ii)	AN	-	2	82	-34+	16	-
90	(i)	AN	TBh	6	43	-4	40	11
	(ii)		TBm	2	218	82	331	14

Footnote: Symbols are as for Table 1. * site numbers in sections (i) and (ii) refer to Figure 2 and site numbers in sections (iii) and (iv) refer to Figure 3. ** deduced from the thermally-demagnetised collection. + components employed in the group mean calculations of Table 3.

and basic intrusions at great depths (8 kb or 25 km, Battey & McRitchie 1973). The most significant local difference has been observed at Spiterstulen in the NE sector of the Massif (Fig. 3) where the NE− components strongly represented in the Liervassbu area 10 km away, are not represented in the host granulites which are dominated instead by a NE+ component ($D = 50°$, $I = 48°$, $\alpha_{95} = 9°$) contrasting with the record in the intrusions ($D = 70°$, $I = -18°$, $\alpha_{95} = 9°$); since there are differences in the magnetic properties of the two suites and only five sites have been occupied, this is likely to be of local significance.

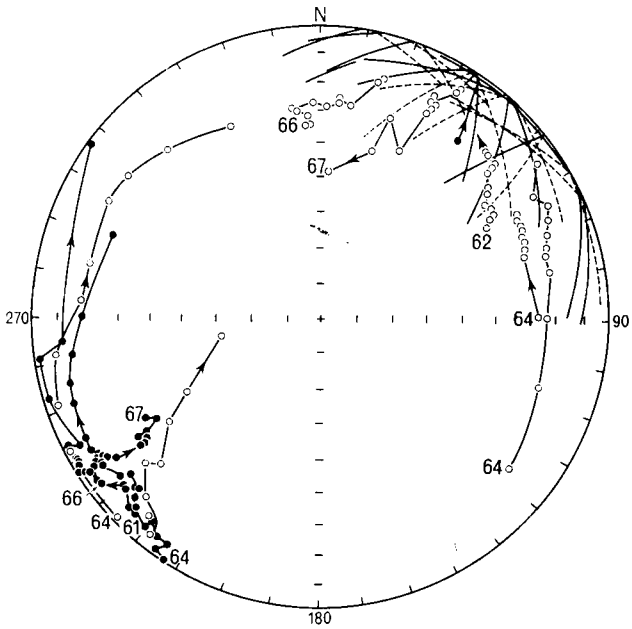


Figure 11. Convergent great circle trajectories described by the magnetism vectors of anorthosite samples from Sognfjord, in the basal part of the Jotun Nappe. Closed symbols are lower hemisphere plots and open symbols are upper hemisphere plots. Stereographic projection.

Table 3. Group mean palaeomagnetic results, metamorphic terranes, south–central Scandinavian Caledonides.

		Mean Directions					Palaeomagnetic Poles			
	No. of sites	D	I	R	k	α_{95}	°N	°E	dp	dm
1. Jotun Nappe, NE-ve/SW+ve components:										
Group 1	11	71	-24	10.9	96	4.7	-2	121	2.7	5.0
Group 2	30	58	-23	29.6	77	3.0	4	132	1.7	3.2
Group 3	28	51	-21	27.7	103	2.7	7	138	1.5	2.8
Group 4	17	43	-22	16.8	88	3.8	10	145	2.1	4.1
2. Jotun Nappe, NE+ve components:										
	23	38	43	21.6	16	7.8	46	136	6.0	9.6
3. Basal Gneiss Complex, NE-ve/SW+ve components:										
	10	42	-24	9.66	37	8.1	9	145	4.6	8.7
4. Basal Gneiss Complex, NE+ve components:										
	19	69	60	18.3	24	7.0	44	91	7.9	10.5

(ii) Whilst NE−/SE+ components are typically resident in narrow and discrete blocking temperature components [Figs 9(a) and (b) and 10], directions are distributed both within sites and between sites in a systematic way: inclinations are largely confined within the range $I = -5$ to $-30°$ whereas declinations range between 30° and 90°. Since rock volumes of the size of individual cores may take protracted intervals of time to cool through their critical blocking temperature ranges at likely plutonic depths of magnetization (see below), they may be expected to have integrated short term variations in the geomagnetic field, and collectively to have recorded long-term changes in the field axis. This inference is partly supported by the recovery of field inversions within individual cores. Within the total sample the contrast between the declinations and inclinations is even more marked with declinations extending through 100° (Fig. 13). Although this kind of distribution can result from small tectonic relative rotations between fault-bounded crustal segments (Macdonald 1980), this explanation is evidently not applicable here for two reasons. Firstly, it is not compatible with the mapped distribution of E–ENE trending faults within the pyroxene granulite massif (Battey & McRitchie 1973) because a large part of the overall distribution is identified in individual tectonic blocks. Secondly, the distribution is sometimes recorded within an individual site or a small number of closely spaced sites. At (ultrabasic) site 31, for example, coherent normally magnetized components range in inclination from −10° to −52° but in declination from 18° to 83°E. Nearby site 34 in the same facies shows a much tighter grouping of normal components with declinations ranging from 22° to 39° and inclinations ranging from −1 to −19°. In the central part of the massif site 53 is an example of a granulite screen between two basic/ultrabasic intrusions. The distribution of components in this screen (declination/inclination, $D/I =$ 40° to 46°/−8 to −15°) is indistinguishable from the closest basic sites 46 ($D/I = 43°$ to 48°/−11 to −29°) and 52 ($D/I = 36°$ to 50°/−4 to −36°) on either side. Although we have conducted no specific studies of magnetic anisotropy, we consider that this effect is unlikely to have influenced the essential features of the directional distributions observed in the Jotun Nappe for two reasons. First, rocks with high degrees of visible mineral foliation were always avoided in the field sampling, and second, the observed palaeomagnetic directions are typically at high angles to the plane containing the regional maximum and intermediate strain axes; this is especially noticeable within the Jotunheimen Massif.

(iii) There is a large-scale regional variation in the magnetic imprint over the areal outcrop of the Jotun Nappe indicative of magnetization over an appreciable interval. To illustrate this point we divide the collection into three groups: the first covers the segment exposed close to sea level along the margins of Sognfjord [(a) in Fig. 13] and comprises results mainly from the anorthosite facies. The other two areas are exposed at levels topographically 1–2 km higher; they comprise the southern half of the Jotunheimen Massif (Fig. 3) incorporating rocks on both sides of the Laerdal–Gjende Fault [(b) in Fig. 13] and the northern part of the Jotunheimen where the bulk of the samples are derived from the Leirvassbu area near to the centre of the Massif. On the basis of the behaviour of the thermally demagnetized samples, the components are classed as of high, medium or low blocking temperature using the same scheme adopted for the BGR. With this classification, the bulk of the components recognized in the Jotun Nappe possess medium blocking temperatures and are subtracted close to the Curie point of magnetite. Only sporadically is there evidence for discrete high blocking temperature components and no coherent groupings of these components emerge. Normal and reversed directions are approximately equally represented in the basal part of the nappe with each group illustrating a compatible distribution and means which are statistically antiparallel ($D/I = 245°/29°$, $\alpha_{95} = 7.9°$ and $D/I = 57°/-22°$, $\alpha_{95} = 7.6°$, calculated from $N = 45$ and 40 components respectively). The southern sector of the Jotunheimen Massif defines a single swathe of magnetic directions embraced by a *circa* 100° distribution of declinations (Fig. 13). Normally magnetized components are much more prominently represented here than reversely magnetized components. [The minor and distinct group of NE+ components in this area (Fig. 12b) is characterized by low unblocking temperatures and has a possible interpretation discussed at the end of this paper]. In the interior of the nappe pile components are almost entirely of normal

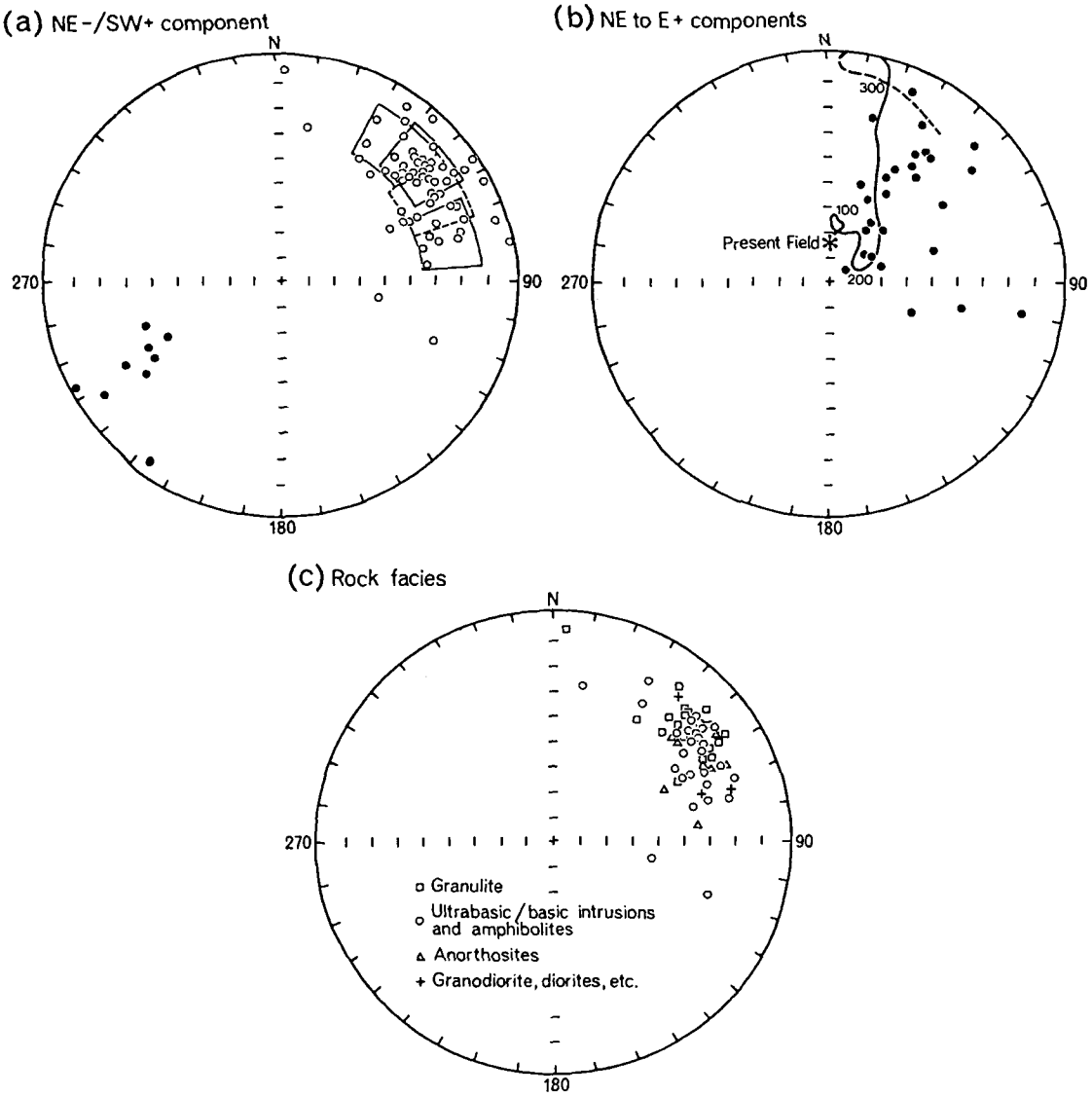


Figure 12. Mean directions of magnetization recorded by two or more cores from the same sampling site separated as (a) those components belonging to the majority population of E to NE−/W to SW+ directions and (b) those components with N–NE+ directions; the latter are compared with the palaofield direction for this region between 310 Ma and the present day predicted from the mean APW path of the Eurasian plate (Piper 1987). (c) Components of magnetization categorized according to the rock lithologies from which they are resolved. Equal area projections. The symbols are as for Fig. 8 and the sectors in (a) define the groups of directions incorporated in the mean calculations of Table 3.

polarity and comprise a more discrete distribution than that found in the southern part of the Massif (Fig. 13c); a similar distribution of declinations and inclinations clusters about a mean of $D = 55°$, $I = -25°$. A minor population of NE+ components is also present here and is essentially unblocked by 500 °C.

(iv) The swathe of medium to high blocking temperature components in the basement are, in terms of their declination, largely discrete from the prominent overprint components recorded in the Devonian sedimentary basins along the west coast of Norway (Torsvik *et al.* 1988). The latter are mostly reversely magnetized and yield a mean of $D/I = 218°/20°$. Rarely components with this direction are observed in the basement; in sample 60-2 shown in Fig. 10, for example, a small residual remanence with southerly shallow direction is identified in haematite following removal of the NE positive component below 570° [see also 10-2 in Fig. 7(a)]. Otherwise only some of the low blocking temperature components identified in the basement rocks could conceivably be linked with the magnetic imprint in these basins, and notably with a 'B' direction of inferred Mesozoic origin yielding a mean of $D/I = 22°/54°$ (Torsvik *et al.* 1988).

From the foregoing discussion and regional comparisons (Section 5.1) we interpret the predominant distribution of declinations recorded in the Jotun Nappe as a consequence of plate rotation during the interval of post-tectonic uplift and cooling of the orogen. The temperature interval recorded embraces at least one inversion of the ancient field and would have been somewhat lower and broader than the range of critical blocking temperature spectra determined in

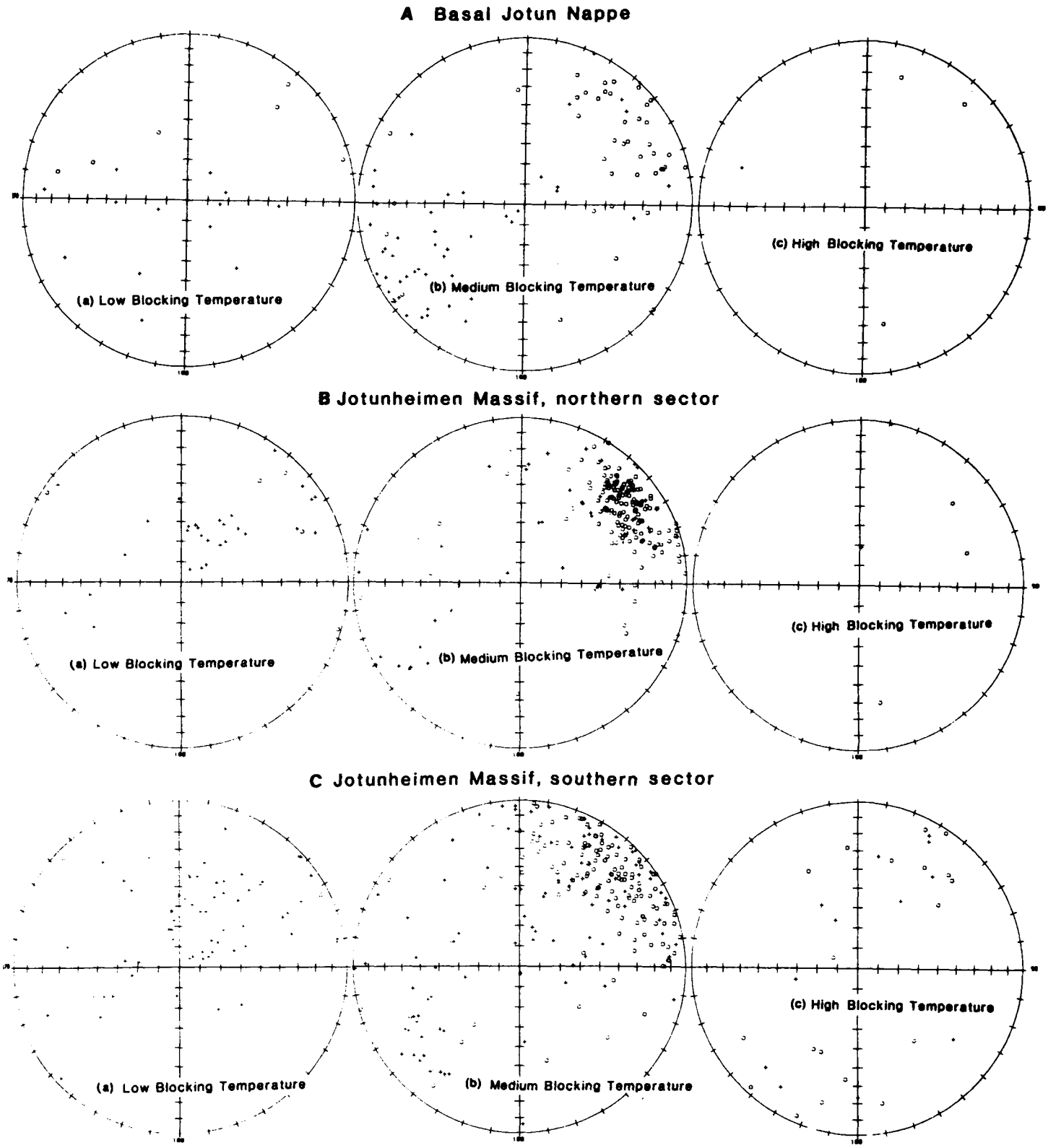


Figure 13. Distributions of all low, medium and high blocking temperature components in the Jotun Nappe grouped as an areal basis from (a) the basal sector of the nappe (NW) exposed close to sea level along Sognfjord, (b) the SE sector of the nappe exposed at similar altitudes mainly in the Tyin–Gjende area (see Figs 1 and 2), and (c) the central part of the nappe complex exposed 1–2 km above sea level in the Jotunheimen Massif. The open circles refer to upper hemisphere plots and the crosses to lower hemisphere plots: equal area projections.

the laboratory (Pullaiah *et al.* 1975; Dodson & McClelland-Brown 1980). This range is typically *circa* 580°–520 °C in the central parts of the Jotunheimen Massif but is, at present, difficult to date absolutely by reference to the geological history. While the Jotun rocks were evidently heated above Ar blocking temperatures, the granulites were remarkably little strained and recrystallized by Caledonian events; visible effects are confined to local retrogression along fracture planes (Twist 1985). Bryhni, Fitch & Miller (1971) report ^{40}Ar:^{39}Ar plateaus of *circa* 400 Ma in the basal Jotun Nappe and 490 Ma in the western part of the BGR. ^{40}Ar:^{39}Ar studies in the Trondalag area 300 km to the north of this study area appear to define nappe translation to higher crustal levels at 430–425 Ma, while Scandian ages in the lower crustal units range from 410 Ma in the central sector to 385 Ma in the western part of the orogen (Dallmeyer 1988). Battey & McRitchie (1973) quote K–Ar ages in the range 475–428 Ma (data of J. G. Mitchell; isolated late Precambrian ages are also recorded) from biotites and feldspars of the Jotunheimen Massif. The authors link these ages to high level emplacement of the pyroxene granulites against the lower grade rocks to the south, and they probably pre-date the magnetization ages because the peak of orogenic activity was reached slightly later in Upper Silurian to Lower Devonian times (Roberts & Gee 1985). Magnetization accompanying cooling following this (Scandian) orogenic pulse is compatible with Sm–Nd and U–Pb ages in the range 425–408 Ma succeeded by Rb–Sr mineral ages on micas which group at around 395 Ma in the eclogitic facies at the western margin of BGR (Griffin *et al.* 1985). Formation of the eclogites by crustal prograde metamorphism can be modelled in terms of a regional NE trending distribution of palaeogeotherms (Fig. 15); it suggests that the BGR was over-ridden by the Laurentian Plate in a continent–continent collision and attained temperatures in the range 500–650 °C (Griffin & Brueckner 1985). These authors attribute Lower–Middle Silurian Sm–Nd ages (428–421 Ma) in this terrane to initial stages of uplift associated with crustal imbrication. The somewhat younger Rb–Sr mineral ages presumably define later stages in the uplift history. Dates in the range 390–386 Ma (Scharer 1980) indicate that this phase may have continued into late Lower Devonian times, in which case it overlapped with deposition of molasse at the western margin of the BGR. Preservation of the molasse basins requires that surface levels were attained by these times (Steel, Siedlecka & Roberts 1985). Completion of uplift is less well constrained in the interior of the Nappe pile although minerals in joints close to the Laerdel–Gjende fault zone (Fig. 3) are dated at 367 Ma (Rb–Sr, Scharer 1980), yielding a minimum age for the transition from brittle to ductile conditions in this area (Milnes & Koestler 1985).

Representative mean directions and pole positions are derived by inverting reversed site directions (Table 2) to their normal equivalents and sub-dividing the resulting distribution into four 20° × 20° splines as shown in Fig. 12(a). The group means are given in Table 3 and define a 28° declination change accompanied by no significant change of inclination. Site mean directions are treated *in situ* because the absence of a Siluro–Devonian reference frame in this plutonic terrane precludes identification of any later deformations which might have influenced these rocks. However, the enormous dimensions of the terrane and their internal consistency imply that deformations post-dating uplift cooling are unlikely to have a significant effect on the mean calculations: a regional tilt of no more than 15° across the 200 km width of this study area would, for example, be expressed as a differential uplift of the full thickness of the crust.

5 DISCUSSION

5.1 Implications to regional and global APW

The comparative APW path from the British Caledonides commencing with results from rocks of Lower Silurian age forms a basis for interpreting the equivalent pole positions and is illustrated in outline form in Fig. 14. This path comprises a westerly extension recorded by Upper Silurian and Lower Devonian rocks and suggested by studies more than three decades ago (Stubbs 1958). The general form has been substantiated by more recent studies since then (Briden *et al.* 1973, 1984). Since geological evidence implies that the Eurasian and Laurentian forelands of the Scandinavian Caledonides were sutured by these times (see for example, Roberts & Gale 1978), it is logical to expect this path to emerge from studies of contemporaneous rocks in continental Europe. This has not yet proved to be the case and possible causes have been extensively discussed (see for example Creer 1968; Storetvedt & Halvorsen 1968; Briden *et al.* 1973). The westerly extremity of the path summarized in Fig. 14 continues a long latitudinal swathe recognized in uplift and cooling magnetizations including results from metamorphic rocks of the Northern Highlands of Scotland (Esang & Piper 1984) and the younger gabbros of Aberdeenshire (Sallomy & Piper 1973; Watts & Briden 1984). This swathe also accommodates results from the 'younger' granite suite indicating that most, if not all, of this record is the signature of a thermochemical event of Silurian age (Torsvik 1985). The concensus of data from this sector of the Caledonides is that the polar location near 350°E, 20°S in Fig. 14 recorded by the 420–415 Ma younger granite suite (e.g. Torsvik, Løvlie & Storetvedt 1983; Torsvik 1985) was occupied prior to a rapid 50°–60° migration to the west by 410–405 Ma in Lower Devonian times. This latter pole position is prominently recorded by primary components in the molasse facies of the orogen (Briden *et al.* 1984; Piper 1987). A single result from the Comrie diorite (408 Ma, Lower Devonian) further suggesting that the APW path could continue west of the one shown in Fig. 14 lies close to the Highland Boundary Fault where local rotation may be responsible for divergence of this pole from the path defined by the remaining data (Turnell 1986).

Results from Middle–Upper Devonian rocks plot SE of the Lower Devonian data (Piper 1979) producing a lobe in the path of Fig. 14. This position is not apparently repeated by the younger palaeofield, but ages and positions of the poles within this group (although relevant to interpretation of the present data) are poorly constrained. In contrast, studies of Lower Carboniferous volcanic rocks in Britain (Wilson & Everitt 1963; Piper, Atkinson & Norris, in preparation) yield poles plotting near the SE extremity of the Siluro–Devonian group and close to the Jotun Group 4 pole (Table 3) from the 'A' magnetizations (Fig. 14). If the

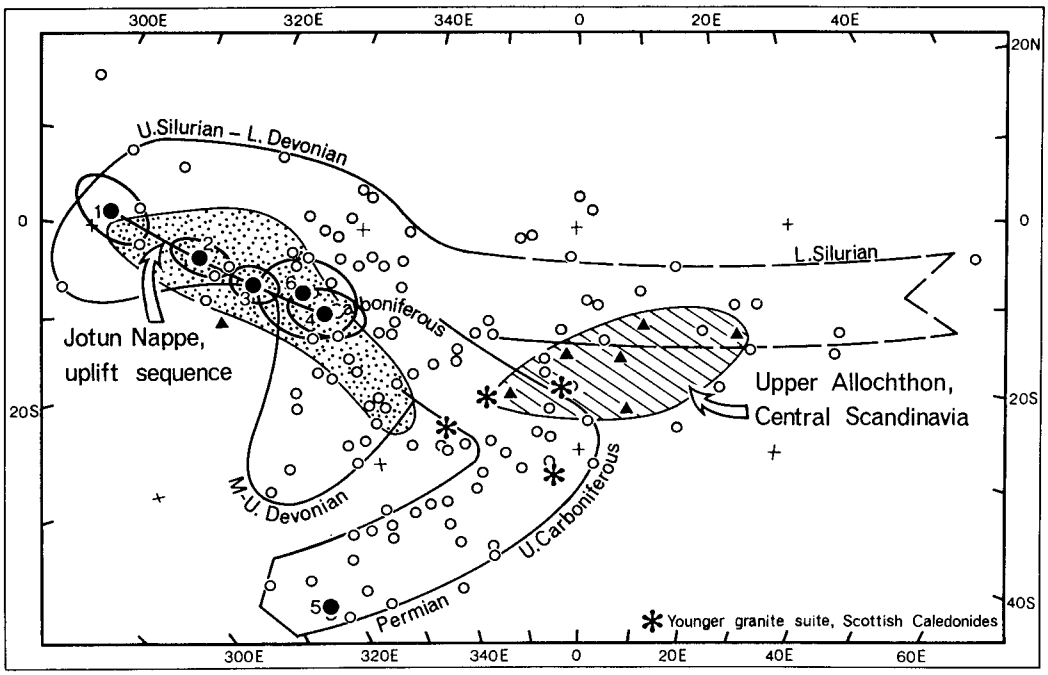
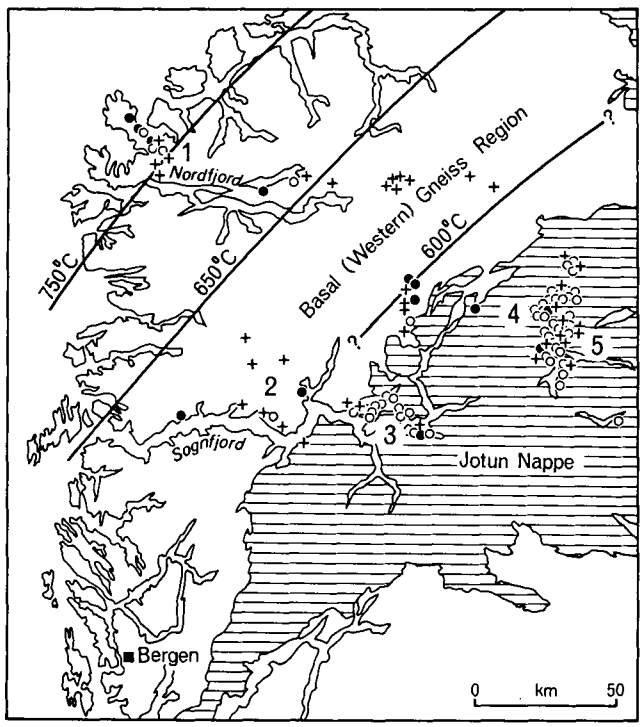


Figure 14. The apparent polar wander path of the south pole between early Silurian and Permian times. The pre-Lower Carboniferous segment of this path is defined largely from British data indicated by open circles and referenced in Piper (1987). The swathe of poles defined from the characteristic remanence record in the Jotun Nappe is shown by the poles 1 to 4 and pole 6 is derived from the NE−/SW+ axis in the Basal Gneiss Region. Pole 5 is the mean of NE+ components in the Jotun Nappe linked to Mesozoic tectonism. The stippled area incorporates the swathe of poles identified in the western part of the orthotectonic Caledonides of Scotland by Watts (1982). The stars are key poles from the younger granite suite in the Scottish Caledonides linked to 420–415 Ma ages (Torsvik 1985) and the shaded area incorporates the most prominent uplift-related remanences recorded by basic bodies in the central–northern Scandinavian Caledonides (Piper *et al.* 1990).

Middle–Upper Devonian group is substantiated by further work, it will require that the APW path subsequently doubled back on itself to execute a loop between Lower Devonian and Lower Carboniferous times. Whilst this part of the Jotun polar swathe could therefore, be as young as Lower Carboniferous in age, this would require that the Middle and Upper Devonian path has gone entirely unrecorded in this terrane. In the context of the geological history, a Silurian to early Devonian age for the Jotun polar swathe (compatible with the radiometric evidence outlined above) is most likely. Along the coastal margin of western Norway deep erosion and truncation of the Finnemarkian and Scandian structures took place prior to rifting and deposition of the Kvamshesten, Hasteinen and Hornelen sediments. The oldest magnetizations in these successions are linked to a tectonic pulse of early Carboniferous age and yield poles plotting at 330–345°E, 10–25°S compatible with the British Lower Carboniferous data (Torsvik *et al.* 1988). They therefore lie SE of the Jotun swathe in Fig. 14. Since the Jotun 'A' magnetizations were most probably acquired at appreciable crustal depths during cooling through the critical blocking temperature ranges prior to this deep erosion, they are reckoned to be considerably older than poles from the late- to post-orogenic molasse facies. Nevertheless, a *general* southeasterly APW motion between early Devonian and Lower Carboniferous times is implied by the collective data.

There is some regional evidence that both the outward and return paths of the Siluro–Devonian APW loop in Fig. 14 are defined within the Jotun Nappe and BGR. In the high structural levels of the Jotun Nappe the declinations are NW (area mean 5 in Fig. 15, equivalent palaeopole at 317°E, 5°N) and they become progressively more westerly towards the base of the nappe, (area means 4 and 3, equivalent palaeopoles at 310°E, 5°N and 310°E, 2°N respectively) and in the higher structural levels of the BGR (area mean 2, equivalent palaeopole at 299°E, 5°S). These declinations yield poles plotting progressively more to the west of the swathe in Fig. 14 implying east to west APW motion. At the deepest exposed zone of the BGR (Fig. 15), which probably includes the levels last cooled through the range of critical blocking temperatures, the declination swings back to a NW quadrant (area mean 1 in Fig. 15, equivalent palaeopole at 313°E, 3°S) and equivalent to a SE migration of the APW path in Fig. 14.

The lower blocking temperature 'B' remanences are most widely recorded in the central part of the study area and beneath the Jotun Nappe. This would appear to be because the rocks here were subjected to much more extensive hydration and concomitant retrogression than those in the Jotun Nappe above. The subsequent magnetic imprint is thus more dependent on the environment during metamorphic retrogression than on the peak metamorphic temperatures attained. Fig. 15 illustrates the importance of the inherited rock facies in the magnetic record of this metamorphic terrane: although the primary metamorphic facies are of Precambrian age, the Caledonian magnetic imprint has been more thoroughly acquired by the higher grade material in the east (where Caledonian metamorphic temperatures were lower) than by the retrogressed material in the west (where Caledonian metamorphic temperatures were higher). The most prominent Caledonian overprint is also in the zone



Area Means

	D	I	α_{95}	N
1	57·3	-35·2	10·8	29
2	72·0	-29·5	11·2	32
3	58·9	-26·3	5·4	85
4	57·3	-20·9	3·0	168
5	49·9	-25·3	4·5	172

Magnetisations:
• Reversed
o Normal
+ E to NE+
—— Caledonian isotherms

Figure 15. The distribution of normal and reversed magnetizations belong to the 'A' (NE−/SE+) group and to the NE to E positive ('B') group of components together with the isotherms deduced from Caledonian mineral assemblages by Griffin & Brueckner (1985). Mean directions of magnetization calculated from samples falling within each areal division numbered 1 to 5 are also listed.

where the data of Griffin *et al.* (1985) suggest that the temperatures attained during Caledonian orogenesis were close to, or somewhat below, the critical magnetic blocking temperature ranges. [The maintenance of these temperatures for several tens of millions of years would have been sufficient to unblock any pre-Caledonian remanence (Pullaiah *et al.* 1975)]. Temperatures well above the blocking temperatures of both magnetite and haematite were attained within the western part of the BGR, but the acquired remanence record subsequently was controlled by the strong hydration during retrogression and orogenic collapse; this presumably took place under mildly reducing conditions because haematite is not widely developed.

The palaeomagnetic poles from the intermediate to high blocking temperature components in the BGR and Jotun Nappe are the first results from continental Europe to plot within the westerly Siluro–Devonian path identified from the British Caledonides. Since this is their most probable age, an important implication of these poles is to show that this feature of APW is not peculiar to Britain but is also applicable to continental Europe and (since the Iapetus suture was by then closed along this sector) to Laurentia as well. Since rocks of this age are widely studied elsewhere and especially in the Maritime Provinces of Canada, the failure to locate it hitherto must reflect the widespread remagnetization of the studied (predominantly sedimentary) rock units of this age. The phenomena of pervasive Upper Palaeozoic remagnetization, and notably by a profound event during the Permo–Carboniferous reversed (PCR) superchron, has been extensively discussed in the literature (see for example Creer 1968; Roy & Morris 1983; Piper 1987) and need not be reiterated here.

Precise correlation of palaeomagnetic results from the Scandinavian Caledonides with the British Caledonides is precluded by Mesozoic and Cainozoic crustal attenuation within the North Sea Basin. This effect is difficult to quantify but seems unlikely to be responsible for rotating the British palaeomagnetic vectors by more than a few degrees anticlockwise with respect to Scandinavia. It is therefore legitimate to compare the uplift and cooling records of the two sectors of the orogen in general terms in present-day coordinates. The long latitudinal APW swathe in Fig. 14 is recognized from the Aberdeenshire sector of the Grampian Highlands (Watts & Briden 1984). Comparable results are forthcoming from the Northern Highlands in the interior part of the orogen immediately to the north of the Great Glen Fault (Esang & Piper 1984). Since these results come from close to the thermal 'core' of the orogen as defined by isotopic closure temperatures and metamorphic facies (Dewey & Pankhurst 1970), it is surprising to find the younger westerly Siluro–Devonian sector of the APW path recognized at the western margin of the orogen (Watts 1982) where lower grade metamorphic facies and old closure ages would otherwise imply an earlier age of acquisition. In Scandinavia the latitudinal segment of the path is recorded in the central part of the orogen (Piper *et al.* 1990 and Fig. 14) while, as the present results have shown, the westerly segment is recorded in part of the southern sector of the orogen. The Jotun Nappe and BGR results in fact correlate very closely with the results of Watts (1982) from the western Scottish Caledonides (stippled area in Fig. 14). Thus the length of this swathe is recorded within the two segments of the orogen, but not in the pattern predicted from simple geometrical considerations of the likely uplift history. When the chronology of this path is more precisely defined, it should enable zones of thermo- and thermochemical remanence acquisition in metamorphic terranes to be isolated, and perhaps for the effects of fluid migration due to orogenic loading (e.g. Oliver 1986) to be recognized.

5.2 Post-Caledonian overprinting events

It remains to consider the significance of the minority population of NE+ components which are mostly subtracted by thermal cleaning up to 500 °C and have directions lying close to the predicted palaeofield migration path derived from the APW path between *circa* 260 and 140 Ma (Fig. 12b). This correlation identifies no single younger overprinting event but suggests instead a range of events broadly of Mesozoic age; whatever their cause, they have clearly had a more profound effect in this area than tectonism in the Oslo Rift to the SE during Permo–Carboniferous times (see, for example, Storetvedt *et al.* 1986) for which the characteristic reversed direction (*circa* $D/I = 200°/-30°$) has gone unrecorded. The overprints are

prominently developed in the vicintiy of the Laerdal–Gjende Fault (Fig. 3 and Table 2), a major ENE–SSW trending lineament responsible for downthrowing retrogressed amphibolite facies terrane of high-level origin against the pyroxene granulite Massif to the north. The fault incorporates a zone of quartzo-feldspathic crush rocks recrystallized during retrogressive metamorphism at epidote–amphibolite grade up to 1.5 km wide (Battey & McRitchie 1973). The fault zone is closely followed by a large negative Bouguer anomaly interpreted as the plunging root zone of the nappe complex (Smithson & Ramberg 1970). It was evidently established along a boundary of Caledonian age and repeatedly reactivated subsequently. These overprints are indirect evidence for Mesozoic movements along this fault zone. Mesozoic overprints have also been identified in the vicinity of the Dalsfjord Fault at Altøy in western Norway which has a comparable trend [Sturt & Torsvik (1986), see also evidence for post-Caledonian fault movements at the coastal margin of the BGR noted by Torsvik *et al.* (1988)].

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REFERENCES

Battey, H. M. & McRitchie, W. D., 1973. The petrology of the pyroxene–granulite facies rocks of Jotunheimen, Norway, *Norsk. Geol. Tidsskr.*, **53,** 237–266.

Briden, J. C., Morris, W. A. & Piper, J. D. A., 1973. Palaeomagnetic studies in the British Caledonides—VI. Regional and Global implications, *Geophys. J. R. astr. Soc.*, **34,** 107–134.

Briden, J. C., Turnell, H. B. & Watts, D. R., 1984. British palaeomagnetism, Iapetus Ocean and the Great Glen Fault, *Geology,* **12,** 428–431.

Brueckner, H. K., 1979. Precambrian ages from the Geiranger–Tafjord area of the Basal Gneiss Region, west Norway, *Norsk. Geol. Tiddsskr.*, **59,** 141–159.

Bryhni, I. & Sturt, B. A., 1985. Caledonides of southwestern Norway, in *The Caledonide Orogen—Scandinavia and Related Areas,* pp. 89–107, eds Sturt, B. A. & Gee, D. G., John Wiley, Chichester.

Bryni, I., Fitch, F. J. & Miller, J. A., 1971. $^{40}Ar/^{39}Ar$ dates from recycled Precambrian rocks in the Gneiss region of the Norwegian Caledonides, *Norsk. Geol. Tidsskr.*, **51,** 391–406.

Creer, K. M., 1968. Palaeozoic palaeomagnetism, *Nature,* **219,** 246–250.

Cuthbert, S. J., Harvey, M. A. & Carswell, D. A., 1983. A tectonic model for the metamorphic evolution of the Basal Gneiss Complex, Western South Norway, *J. metamorphic Geol.*, **1,** 63–90.

Dallmeyer, R. D., 1988. Polyphase tectonothermal evolution of the Scandinavian Caledonides, in *The Caledonian–Appalachian Orogen,* pp. 365–379, eds Harris, A. L. & Fettes, D. J., Geological Society Special Publication No. 38, Blackwell Scientific Publications, Oxford.

Dewey, J. F. & Pankhurst, R. J., 1970. The evolution of the Scottish Caledonides in relation to their isotopic age pattern, *Trans. R. Soc. Edin.*, **58,** 361–389.

Dodson, M. H. & McClelland-Brown, E., 1980. Magnetic blocking temperatures of single-domain grains during slow cooling, *J. geophys. Res.*, **85,** 2625–2637.

Esang, C. B. & Piper, J. D. A., 1984. Palaeomagnetism of Caledonian intrusive suites in the Northern Highlands of Scotland: constraints to tectonic movements within the Caledonian orogenic belt, *Tectonophysics,* **104,** 1–34.

Gebauer, D., Lappin, M. A., Grunenfelder, M. & Wyttenbach, A., 1985. The age and origin of some Norwegian eclogites: A U–Pb zircon and REE study, *Chem. Geol.*, **52,** 2247–247.

Griffin, W. L. & Brueckner, H. K., 1985. REE, Rb–Sr and Sm–Nd studies of Norwegian eclogites, *Chem. Geol.* (Isotope Geoscience Section), **52,** 249–271.

Griffin, W. L., Austrheim, K., Brastas, K., Bryhni, I., Krill, A. G., Krogh, E. J., Morl, M. B. E., Qvale, H. & Torudbakken, B., 1985. High pressure metamorphism in Scandinavian Caledonides, in *The Caledonide Orogen–Scandinavia and Related Areas,* pp. 783–802, eds Gee, D. G. & Sturt, B. A., John Wiley, Chichester.

Holtendahl, O., 1938. Trekk av det skandinaviske fjell kjedestroks historie, *Skand. Naturforsk. mote,* **19,** Helsingfors, 129–145.

Lappin, M. A., 1966. The field relationships of basic and ultrabasic masses in the basal gneiss complex of Stadlandet and Almklovdalen, Nordfjord, Southwestern Norway, *Norsk. Geol. Tiddsskr.*, **46,** 439–496.

Lappin, M. A., 1974. Eclogites from the Sunndel–Grubse Ultramafic mass, Almklovdalen, Norway and the P–T history of the Almklovdalen mass, *J. Petrol.*, **18,** 567–601.

Lappin, M. A., Pidgeon, R. T. & Van Breemen, O., 1979. Geochronology of basal gneisses and mangerite syenites of Stadlandet, west Norway, *Norsk. Geol. Tiddsskr.*, **59,** 161–181.

Løvlie, R. & Mitchell, J. G., 1982. Complete remagnetisation of some Permian dykes from Western Norway induced during burial/uplift, *Phys. Earth planet. Inter.*, **30,** 415–421.

Macdonald, W. D., 1980. Net tectonic rotation, apparent tectonic rotation and the structural tilt correction in palaeomagnetic studies, *J. geophys. Res.*, **85,** 3659–3669.

McWilliams, M. O. & Dunlop, D. J., 1978. Grenville palaeomagnetism and tectonics, *Can. J. Earth Sci.*, **15,** 587–695.

Milnes, A. G. & Koestler, A. G., 1985. Geological structure of Jotunheimen, southern Norway (Sognefjell–Valdres cross section), in *The Caledonide Orogen–Scandinavia and Related Areas,* pp. 457–474, eds Gee, D. G. & Sturt, B. A., John Wiley, Chichester.

Morgan, G. E., 1981. Palaeomagnetism of a slowly-cooled plutonic terrane in western Greenland, *Nature,* **259,** 383–385.

Oliver, J., 1986. Fluids expelled tectonically from orogenic belts: their role in hydrocarbon migration and other geological phenomena, *Geology,* **14,** 99–102.

Piper, J. D. A., 1979. Aspects of Caledonian palaeomagnetism and their tectonic implications, *Earth planet. Sci. Lett.*, **46,** 443–461.

Piper, J. D. A., 1981. The altitude dependence of magnetic remanence in the slowly-cooled Precambrian plutonic terrain of west Greenland, *Earth planet. Sci. Lett.*, **54,** 449–466.

Piper, J. D. A., 1985. Palaeomagnetic study of the Nagssugtoqidian mobile belt in central–west Greenland, *Precambrian Res.*, **28,** 75–110.

Piper, J. D. A., 1987. *Palaeomagnetism and the Continental Crust,* Open University Press, Milton Keynes.

Piper, J. D. A., Poppleton, T. J., Mason, R. & Griffiths, A. J., 1990. The uplift magnetisation record in the central Scandinavian Caledonides: results from basic igneous complexes, *Tectonophysics,* in press.

Poppleton, T. J. & Piper, J. D. A., 1990. Palaeomagnetism of a Caledonian ophiolite, Karmøy, SW Norway, *Phys. Earth planet. Inter.*, in press.

Pullaiah, G., Irving, E., Buchan, K. L. & Dunlop, D. J., 1975. Magnetisation changes cuased by burial and uplift, *Earth planet. Sci. Lett.*, **28,** 133–143.

Roberts, D. & Gale, G. H., 1978. The Caledonian–Appalachian Iapetus Ocean, in *Evolution of the Earth's Crust*, pp. 255–342, ed. Tarling, D. H., Academic Press, London.

Roberts, D. & Gee, D. G., 1985. An introduction to the structure of the Scandinavian Caledonides, in *The Caledonian Orogen–Scandinavia and Related Areas*, pp. 55–68, eds Gee, D. G. & Sturt, B. A., John Wiley, Chichester.

Roy, J. L. & Morris, W. A., 1983. A review of palaeomagnetic results from the Carboniferous of North America; the concept of Carboniferous geomagnetic field horizon markers, *Earth planet. Sci. Lett.*, **65,** 167–181.

Rumble, D., 1978. Oxide minerals in metamorphic rocks, in *Oxide Minerals*, vol. 3, pp. R1–R24, ed. Rumble, D., Mineral Society of America Short Course Notes, Washington, DC.

Sallomy, J. T. & Piper, J. D. A., 1973. Palaeomagnetic studies in the British Caledonides IV—Lower Devonian lavas of the Strathmore Region, Scotland, *Geophys. J. R. astr. Soc.*, **34,** 49–68.

Scharer, U., 1980. U–Pb and Rb–Sr dating of a polymetamorphic nappe terrain: the Caledonian Jotun nappe, Southern Norway, *Earth planet Sci. Lett.*, **49,** 205–218.

Senenayake, W. E. & McElhinny, M. W., 1981. Hysteresis and susceptibility characteristics of magnetite and titanomagnetites: interpretation of results from basaltic rocks, *Phys. Earth planet. Inter.*, **26,** 47–55.

Smithson, S. B. & Ramberg, I. B., 1970. Geophysical profile bearing on the origin of the Jotun nappe in the Norwegian Caledonides, *Geol. Soc. Am. Bull.*, **81,** 1571–1576.

Stacey, F. D., 1963. The physical theory of rock magnetism, *Phil. Mag. Supp. Adv. Phys.*, **12,** 46–133.

Steel, R., Siedlecka, A. & Roberts, D., 1985. The Old Red Sandstone basins of Norway and their deformation: a review, in *The Caledonide Orogen–Scandinavia and Related Areas*, pp. 293–316, eds Gee, D. G. & Sturt, B. A., John Wiley, Chichester.

Storetvedt, K. M. & Halvorsen, E., 1968. On the palaeomagnetic reliability of the Scottish Devonian lavas, *Tectonophysics*, **5,** 447–457.

Storetvedt, K. M. & Torsvik, T. H., 1983. Palaeomagnetic re-examination of the basal Caithness Old Red Sandstones: aspects of local and regional tectonics, *Tectonophysics*, **98,** 151–164.

Storetvedt, K. M., Carmichael, C. M., Hayatsu, A. & Palmer, H. C., 1978. Palaeomagnetism and K/Ar results from the Duncansby volcanic neck, NE Scotland: superimposed magnetisations, age of igneous activity and tectonic implications, *Phys. Earth planet. Inter.*, **16,** 379–392.

Storetvedt, K. M., Pedersen, S., Løvlie, R. & Halvorsen, E., 1986. Palaeomagnetism in the Oslo Rift Zone, in *Tectonics and Geophysics of Continental Rifts*, pp. 289–296, eds Ramberg, I. B. & Neumann, E. R., Reidel, Dordrecht.

Stubbs, P. H. S., 1958. Continental drift and polar wandering: a palaeomagnetic study of British and European Trias and the British Old Red Sandstones, *PhD thesis*, University of London.

Sturt, B. A. & Torsvik, T. H., 1986. *Palaeomagnetism and Dating of Fault Movements* (abstract), 17th Nordic Geology Meeting, Helsinki.

Taylor, G. K., 1988. A palaeomagnetic study of a Caledonian ophiolite, *Geophys. J.*, **94,** 157–166.

Torsvik, T. H., 1985. Palaeomagnetic results from the Peterhead granite, Scotland; implications for regional late Caledonian magnetic overprinting, *Phys. Earth planet. Inter.*, **39,** 108–117.

Torsvik, T. H., Løvlie, R. & Storetvedt, K. M., 1983. Multi-component magnetisation in the Helmsdale granite, North Scotland; geotectonic implications, *Tectonophysics*, **98,** 111–129.

Torsvik, T. H., Sturt, B. A., Ramsay, D. M., Bering, D. & Fluge, P. R., 1988. Palaeomagnetism, magnetic fabrics and the structural style of the Hornelen Old Red Sandstone, Western Norway, *J. geol. Soc. Lond.*, **145,** 413–430.

Trench, A., Bluck, B. J. & Watts, D. R., 1988. Palaeomagnetic studies in the Ballantrae ophiolite, southwest Scotland: magnetotectonic and regional tectonic implications, *Earth planet. Sci. Lett.*, **90,** 431–448.

Turnell, H. B., 1986. Palaeomagnetism and Rb–Sr ages of the Ratagan and Comrie intrusions, *Geophys. J. R. astr. Soc.*, **83,** 363–385.

Turner, P., Vaughan, D. J. & Tarling, D. H., 1978. Palaeomagnetic and mineralogical studies of Devonian lactrustine sediments from Caithness, Scotland, *Phys. Earth planet. Inter.*, **16,** 73–83.

Twist, D., 1985. The structural history of the northwestern margin of the Jotunheimen Massif, *Norsk. Geol. Tiddsskr.*, **65,** 151–165.

Watts, D. R., 1982. A multi-component, dual-polarity palaeomagnetic regional overprint from the Moine of northwest Scotland, *Earth planet. Sci. Lett.*, **61,** 190–198.

Watts, D. R., 1985. Palaeomagnetic resetting in the Barrovian zones of Scotland and its relationship to the late structural history, *Earth planet. Sci. Lett.*, **75,** 258–264.

Watts, D. R. & Briden, J. C., 1984. Palaeomagnetic signature of slow post-orogenic cooling of the north-west Highlands of Scotland recorded in the Newer Gabbros of Aberdeenshire, *Geophys. J. R. astr. Soc.*, **77,** 775–778.

Wilson, R. L. & Everitt, C. W. F., 1963. Thermal demagnetisation of some Carboniferous lavas for palaeomagnetic purposes, *Geophys. J. R. astr. Soc.*, **8,** 149–164.